SRI LANKA INSTITUTE OF INFORMATION TECHNOLOGY

Faculty of Computing · Department of Information Technology

SOS EMERGENCY SAFETY SYSTEM: AN AI-BASED TOURIST EMERGENCY DETECTION AND POLICE RESPONSE PLATFORM

Individual Component of the Research Project R26-IT-152

TourCeylon: An Integrated AI-Driven Platform for Tourist Safety, Itinerary Planning, Wellness Matching, and Crowd Prediction in Sri Lanka

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IT22108654

Bachelor of Science (Honours) in Information Technology
Specialising in Information Technology

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Dissertation submitted in partial fulfilment of the requirements for the Bachelor of Science (Honours) degree in Information Technology specialising in Information Technology

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October 2026

DECLARATION

I declare that this is my own work and this dissertation does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or institute of higher learning and to the best of my knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The above candidate has carried out research for the bachelor's dissertation under my supervision.

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Mr. S. M. B. Harshanath

Signature of the co-supervisor: …… Date: ……

Prof. Samantha Rajapaksha

ABSTRACT

Tourists in Sri Lanka who face theft, a medical emergency, harassment, fire or a road accident must reach the right emergency channel, explain where they are and describe what has happened, often in a second language and under stress. Existing voice hotlines are not linked to the tourist's live location, profile or trip context. This thesis presents the SOS Emergency Safety System, the individual component of the TourCeylon platform (R26-IT-152). An authenticated tourist raises a GPS-anchored SOS by one tap, by text or by voice. The statement is cleaned, converted into TF-IDF features and classified by two Random Forest models, one for the incident type (14 classes) and one for the responder (police, medical or fire). The alert is routed to the nearest station, pushed in real time to police and immigration dashboards with a structured briefing, and queued offline if there is no connectivity. A keyword classifier is the runtime fallback, so an alert is never lost when the AI fails.

Following design science research, the models were trained on a corpus of 264 synthetic, labelled statements and evaluated on a stratified hold-out set against keyword and dummy baselines. The responder Random Forest reached an accuracy of about 0.811 and a macro-F1 of about 0.684. On the fourteen-class task the Random Forest reached a macro-F1 of about 0.359, below the keyword baseline (about 0.496) and Logistic Regression (about 0.707), because of sparse per-class data. The thesis reports this openly and recommends corpus expansion. Language coverage beyond translated input, response time and usability scores were not measured. The work concludes that structured, map-backed digital SOS with machine-learning triage complements national voice hotlines.

Keywords: tourist safety; emergency detection; text classification; Random Forest; multilingual speech; police dashboard

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
API	Application Programming Interface
DSR	Design Science Research
CV	Cross-Validation
DFD	Data Flow Diagram
F1	Harmonic mean of precision and recall
GPS	Global Positioning System
JWT	JSON Web Token
LR	Logistic Regression
MFA	Multi-Factor Authentication
ML	Machine Learning
NER	Named Entity Recognition
NLP	Natural Language Processing
PDPA	Personal Data Protection Act
PII	Personally Identifiable Information
REST	Representational State Transfer
RBAC	Role-Based Access Control
RF	Random Forest
SDG	Sustainable Development Goal
SLTDA	Sri Lanka Tourism Development Authority
SMS	Short Message Service
SOS	Distress alert or emergency help request
STT	Speech-to-Text
SUS	System Usability Scale
TF-IDF	Term Frequency–Inverse Document Frequency
TLS	Transport Layer Security
WER	Word Error Rate

1. INTRODUCTION

This chapter introduces the research context of the SOS Emergency Safety System, which is the individual component of the TourCeylon platform that deals with tourist safety and emergency response in Sri Lanka. It reviews the background and the literature, identifies the research gap, states the research problem and presents the objectives and research questions that guide the remainder of the thesis.

1.1 Background and Literature Survey

The survey below is organised around the functions that an emergency response platform for tourists must perform: understanding the destination context, raising an alarm, understanding what the tourist says, communicating across languages, locating the tourist and the nearest responder, and protecting the personal data involved. Sources include peer-reviewed journal and conference papers, standards, official government and agency publications, and a small number of news reports where no official publication was available. News reports are identified as such and should be replaced by primary statistics where these can be obtained.

1.1.1 Tourism and safety context in Sri Lanka

Perceived safety is one of the factors that influence where tourists choose to travel, and visitors are particularly vulnerable because they lack local knowledge, local language skills and social networks [1], [2]. For Sri Lanka, tourism is a priority sector. A report based on figures from the Ministry of Tourism states that the country received more than 2.36 million international visitors in 2025, a record, and that the government expects at least 3 million in 2026; the same report notes that the average daily spend per tourist was revised to USD 148 [3]. Growth of this kind increases the number of visitors who may need help from police, ambulance and other services in a country whose emergency arrangements and languages are unfamiliar to them.

Several channels already exist. The Sri Lanka Tourism Development Authority (SLTDA) operates the 1912 tourism hotline, which provides travel information, support during unexpected disasters and connection to the Tourism Police Division for complaints about harassment or service quality; the SLTDA page also lists the 1990

ambulance service (Suwa Seriya) for tourists [4]. In October 2024 a further hotline, 1997, was reported as active for tourists to raise security concerns, together with plans to strengthen the Tourist Police Division [5]. In December 2022 the Minister of Tourism announced a planned mobile application through which tourists could check their location and obtain instructions if lost; the source does not confirm that it was launched [6]. These channels are voice-based or informational. They require the tourist to know which number to call and to describe the problem in a language the operator understands.

Open sources show that incidents of this kind are reported in routine tourism settings. A traveller's trip report from March 2025 describes first-aid assistance after a tuk-tuk crash [7]. The Sunday Times reported the testimony of a foreign visitor whose hired car was intercepted and attacked by a group of tuk-tuk drivers [8]. News reports describe a violent taxi robbery in Arugam Bay [9] and the robbery of a Spanish tourist on a train route [10], and 2025 harassment cases involving Dutch and New Zealand tourists led to arrests after video evidence spread online [11], [12]. The Morning quotes tourism and Tourist Police officials on stationing units at popular sites and encouraging prompt reporting [13], and the Sri Lanka Police remind the public that immediate emergencies should use 119 and that misuse of the number blocks genuine callers [14]. Table 1.1 summarises these sources. They are news and forum reports. They show that such incidents are reported and are used in this thesis as problem evidence only, not as statistics on how often incidents occur.

Table 1.1: Public reports of tourism-related safety and emergency problems in Sri Lanka

Source	Reported issue	Relevance to the SOS component
Traveller trip report, 2025 [7]	Tuk-tuk crash injury; first aid	Transport and medical incidents during normal sightseeing
Sunday Times, 2025 [8]	Tuk-tuk group attacks hired car	Harassment and unsafe location; need for a trusted reporting channel
Morning Telegraph, 2025 [10]	Tourist assaulted on a train; phone stolen	Theft and violence on scenic routes
Israel National News, 2025 [9]	Violent taxi robbery near Arugam Bay	Theft; remote coastal areas
Daily Mirror and Ceylon Today, 2025 [11], [12]	Harassment cases; arrests after viral video	Harassment; value of fast digital evidence and reporting

Source	Reported issue	Relevance to the SOS component
SLTDA and Sri Lanka Police [4], [14]	1912 tourism hotline; 119 for immediate emergencies, misuse warning	Official voice channels exist but are not linked to the tourist's live map position or profile
The Morning, 2025 [13]	Tourist Police at popular sites; prompt reporting encouraged	Policy support for faster, structured reporting

> **Evidence note.** The project proposal submitted to the Sri Lanka Police reports that more than 200 tourist fatalities occur each year because of road accidents, harassment and delayed emergency response, and that no app-based emergency dispatch exists for tourists. These statements are taken from the proposal. These figures are claims made in the proposal and are reported here as such.

1.1.2 Tourist safety technologies and SOS mechanisms

Consumer smartphones now include built-in emergency features. The documentation for Google Pixel phones describes an Emergency SOS function that is started by pressing the power button several times, calls the local emergency number, shares the user's location with chosen emergency contacts and can start video recording; the same documentation lists limitations, such as the need for a SIM or eSIM, no operation in airplane mode or with Battery Saver on, and availability of location sharing only in certain countries and account types [15]. Features of this kind demonstrate that a quick trigger and location sharing are accepted design elements. However, they place a voice call to a local number at the centre of the interaction, and the documentation reviewed does not describe interpretation of the content of the request, suggestion of the responder type, or transmission of a verified tourist profile to a police dashboard.

The peer-reviewed literature on integrated tourist emergency systems is considerably thinner than the literature on tourism safety as a management topic. This thesis therefore treats the absence of a documented, integrated tourist SOS platform for Sri Lanka as a gap that is supported by the sources reviewed, and not as a universal claim.

1.1.3 Classification of emergency and crisis messages

Text produced during emergencies has been studied extensively in crisis informatics. Castillo [16] gives a broad treatment of how social-media data are used in disasters and time-critical situations. Imran et al. [17] surveyed methods for processing social-media messages during mass emergencies and identified classification of messages

into information types, for example reports of injured people, infrastructure damage or requests for help, as a core task that is usually solved with supervised learning. The same group released human-annotated Twitter corpora for crisis-related messages and showed that classifiers on simple n-gram features provide solid baselines [18]. Olteanu et al. [19] built CrisisLex, a lexicon of crisis-related terms, and showed that curated keyword lists are useful for collecting relevant messages but have limited coverage on their own, which motivates learned classifiers. Sakaki et al. [20] treated Twitter users as social sensors for real-time event detection, which illustrates the value of rapid automatic interpretation of short texts.

These studies share two characteristics. They address public posts that are written during large events such as earthquakes and floods, and their aim is situational awareness for many observers rather than help for one individual. A tourist who sends a short statement to an SOS system is in a different position: the message is addressed to responders, it is typically a few words long, it may be written under stress, and a wrong decision can send the wrong service to the wrong place. The methods from crisis informatics are therefore relevant, but the data, the labels and the operational context differ.

The use of machine learning inside emergency dispatch has also been tested in practice. A randomised clinical trial of a machine-learning tool that analysed emergency calls to alert dispatchers to possible cardiac arrest reported no significant improvement in the dispatchers' recognition of cardiac arrest [21]. The relevance for this thesis is methodological: results obtained in laboratory evaluation do not automatically translate into operational benefit, which is why the proposed system keeps the human officer in the loop and why the evaluation plan includes usability and field testing in addition to classification metrics.

1.1.4 Classical and transformer-based text classifiers

Automated text categorisation assigns documents to predefined classes using supervised learning [22]. In the vector-space model, text is represented with term frequency–inverse document frequency (TF-IDF) weights, which emphasise terms that are frequent in a document but rare across the collection [23]. Linear classifiers

such as support vector machines and logistic regression are strong baselines for text because the feature space is high-dimensional and sparse [24], and Wang and Manning [25] showed that simple bigram-based linear models are very competitive for short-text tasks. A random forest is an ensemble of decision trees trained on bootstrap samples with random feature subsets [26]; a large comparison of 179 classifiers on 121 data sets found random forests among the best performing families [27]. The implementation of both model families in the scikit-learn library makes them reproducible with modest effort [28].

Transformer-based language models such as BERT [29] reach higher accuracy on many benchmarks but require fine-tuning data, memory and computation, and they are less transparent. Smaller variants such as DistilBERT [30] reduce the cost, and multilingual models such as multilingual BERT [31] and XLM-R [32] support cross-lingual transfer, while Sentence-BERT [33] produces sentence embeddings that can be used with simpler classifiers. These models are important future directions for multilingual emergency text, but with a small, custom corpus they carry a risk of over-fitting and they complicate deployment. For this reason the comparison in this thesis is restricted to random forest and logistic regression on TF-IDF features, with transformer models positioned as future work.

Evaluation under class imbalance requires care. Emergency data typically contain some incident types that are much more frequent than others, and accuracy alone can be misleading [34]. Macro-averaged precision, recall and F1-score, together with a confusion matrix, give a more balanced picture [35]. With small data sets, stratified cross-validation provides more stable estimates than a single split [36], and paired tests such as McNemar's test [37] are recommended for comparing two classifiers on the same test items [38]. Local explanation methods can help to check that a classifier relies on meaningful words [39].

1.1.5 Multilingual speech and translation in emergency communication

Tourists speak many languages, and a distressed person may not be able to speak English. Speech recognition has been advanced by large multilingual models. Whisper was trained on 680,000 hours of multilingual and multitask supervised data collected

from the web and was reported to approach human robustness in English speech recognition without dataset-specific fine-tuning [40]. Neural machine translation, as exemplified by Google's neural machine translation system [41], is widely available through commercial services such as the Google Cloud Translation API [42]. The project proposal selects Whisper for speech-to-text and the Cloud Translation API for translating transcripts to English for police officers.

Reliance on automatic transcription and translation in a safety-critical setting carries risks that are not evident from average accuracy figures: a single mistranscribed word can change the meaning of a statement, low-resource languages such as Sinhala and Tamil generally have less training data than major European languages, and accents, noise and emotional speech degrade recognition. Three design principles follow: the original-language transcript must always be retained and visible to the officer; a language-independent trigger must exist so that an emergency can be reported without speaking; and the machine output must be advisory and open to human correction. The proposed system follows these principles (Section 2.5).

1.1.6 Location-based dispatch and police-side coordination

A distress call is useful only if responders know where the caller is. Browser and smartphone geolocation provide coordinates whose accuracy varies with the device, signal and environment, so the system must show the position together with its source and time and must allow fallback to the last known or manually entered location. Given the tourist's coordinates and a table of facility coordinates, the great-circle distance can be computed with the haversine formula [43] and used to select the nearest suitable station or hospital. The result must be presented on a dashboard that lets an officer acknowledge, escalate and update the incident, and that records response times so that performance can be reviewed. In Sri Lanka, the Tourist Police Division is the natural recipient for incidents involving tourists, and its strengthening has been reported [5].

1.1.7 Security, privacy and data protection

A tourist emergency system handles identity documents, telephone numbers, emergency contacts, location and free-text descriptions of incidents. In Sri Lanka, the Personal Data Protection Act No. 9 of 2022 [44] establishes obligations for processing

personal data. The Data Protection Authority describes the Act as the first of its kind in South Asia, drawing on the European GDPR and applying to processing within Sri Lanka, and reports that it is being phased in, with the Authority's establishment from July 2023 and the remaining parts scheduled to take effect in March 2025 [45].

Technical guidance is available from recognised sources. NIST SP 800-63B describes authentication and lifecycle management and recommends multi-factor authentication for higher assurance levels [46]. JSON Web Tokens are a standard way to carry signed claims between client and server [47], and TLS 1.3 protects data in transit [48]. The OWASP Top 10 lists the most critical web application security risks and provides a checklist for testing [49], and ISO/IEC 27001 offers a management framework for information security [50]. These are used in Section 2.8 as reference points; the thesis does not claim formal certification.

1.1.8 Existing systems and comparison

Table 1.2 compares categories of existing solutions with the proposed system. A cell states what the cited documentation describes; where the sources reviewed do not describe a capability, the cell reads “Not described” and does not assert that the capability is absent. The proposed system column distinguishes capabilities that are part of the current build from those that are only proposed.

Table 1.2: Comparison of existing solutions with the proposed SOS system

Capability	**National hotlines (119, 1990, 1912, 1997)**	**Smartphone emergency SOS (Pixel example)**	**Announced tourist app (2022)**	**Crisis-text research**	**Proposed SOS system**
Verified tourist profile with the alert	Not described	Not described	Not described	Not applicable	Yes: shared TourCeylon login; profile attached to the briefing
One-tap or quick trigger	Voice call	Yes (power button presses)	Not described	Not applicable	One-tap SOS button with mandatory GPS
Location sharing	Verbal description	Yes, to contacts and responders where available	Location check described	Geotagging of some posts	Browser GPS required; live updates; Safe Journey history
Understanding free-text statements	Human operator	Not described	Not described	Yes (supervised classifiers)	TF-IDF with Random Forest (and Logistic Regression comparison); results in Chapter 3

Capability	National hotlines (119, 1990, 1912, 1997)	Smartphone emergency SOS (Pixel example)	Announced tourist app (2022)	Crisis-text research	Proposed SOS system
Suggests responder type	Caller chooses number	Not described	Not described	No	Second Random Forest (police, medical, fire); results in Chapter 3
Multilingual voice input	Operator-dependent	Not described	Not described	Not applicable	Voice path with speech-to-text and translation to English implemented; 10-language target and per-language accuracy not validated
Police-side dashboard and tracking	Internal systems	Not described	Not described	No	Police dashboard with real-time updates and chat briefing; read-only immigration view
Nearest station or facility	Operator knowledge	Not described	Not described	No	Haversine search on stored station coordinates
Works without network	Not applicable	Not described	Not described	Not applicable	Offline queue with replay when online

Source: Compiled by the author from [4], [5], [6], [15], [17] and the project proposal.

1.1.9 Critical synthesis

Taken together, the sources show that (i) tourist safety matters for Sri Lanka and several voice-based channels exist; (ii) consumer devices provide quick triggers and location sharing but are not designed around police response to foreign visitors; (iii) crisis informatics supplies mature methods for classifying emergency text but for public posts rather than individual requests; (iv) speech recognition and machine translation have advanced enough to be considered for multilingual input but carry risks that call for human oversight; and (v) the Sri Lankan data protection regime requires that personal data are handled with care. What the reviewed sources do not show is an integrated, evaluated system for Sri Lanka that combines a verified tourist profile, a quick trigger, classification of the tourist's statement, location-based routing to the nearest station and a police dashboard. Figure 1.1 summarises how the proposed system combines these elements.

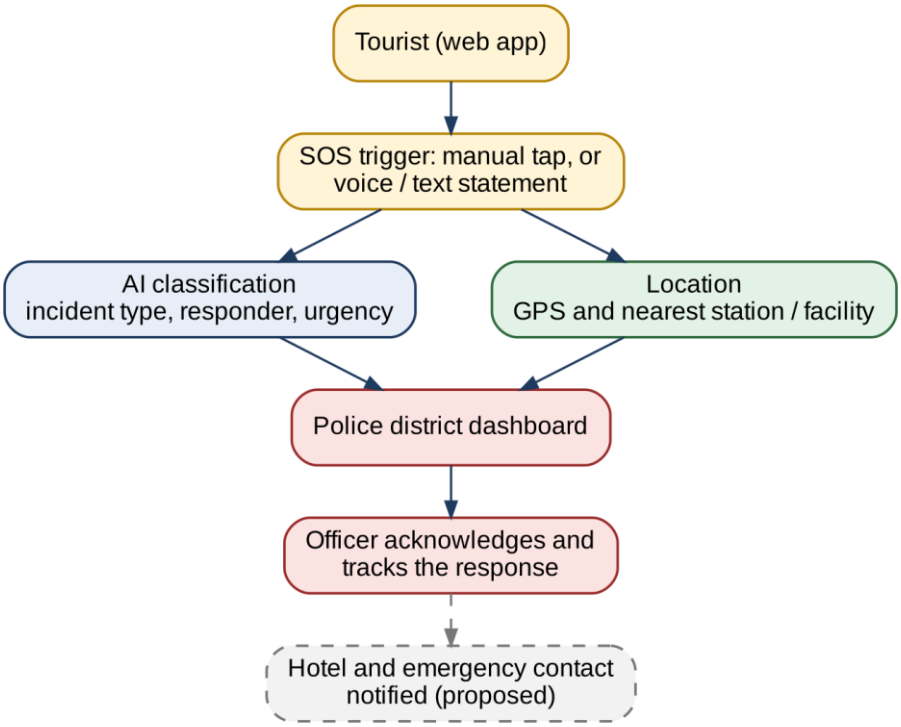


Figure 1.1: Conceptual overview of the proposed SOS Emergency Safety System

1.2 Research Gap

The research gap is formulated from the synthesis above and is limited to what the reviewed sources support. Table 1.3 lists five gaps, what is already available, what remains open and how this research responds. Each gap is accompanied by the status of the evidence that supports it.

Table 1.3: Research gaps, current coverage and the response of this research

No.	Gap	What is already available	What remains open	Response of this research	Evidence status
G1	Classification of individual tourist distress statements for responder routing	Classifiers for crisis posts on social media [17], [18]	Models and labelled data for short statements addressed directly to responders, mapped to police, medical and fire channels	Random Forest incident-type and responder classifiers compared with logistic regression	Supported by the reviewed literature
G2	Integrated tourist-to-police workflow for Sri Lanka	Hotlines, quick-trigger phone features, announced tourist app [4], [15], [6]	A documented, evaluated platform linking verified profile, GPS, classification and a police dashboard	Tourist web app, SOS pipeline, police and immigration dashboards, offline queue and shared TourCeylon login	Supported for the sources reviewed; wider search recommended

No.	Gap	What is already available	What remains open	Response of this research	Evidence status
G3	Multilingual distress input for tourists, including Sinhala and Tamil	Large multilingual speech and translation models [40], [41]	Evidence of use in tourist emergencies in Sri Lanka, with human oversight and per-language evaluation	Optional speech-to-text and translation to English before classification; one-tap SOS as language-independent trigger	Partly supported; per-language results are reported in Chapter 3
G4	Small-data model comparison in this domain	General evidence on linear models and forests [25], [27]	A reproducible RF versus LR comparison on tourist emergency statements, with baselines	Hold-out comparison of Random Forest, Logistic Regression, keyword rules and a dummy baseline on the same split	Addressed by the evaluation in Chapter 3
G5	Privacy-aware design for tourist emergency data under the Sri Lankan Act	General security standards and the Act [44], [46], [48]	Applied design: consent at SOS time, minimum personal data, role-based access, consented-only immigration view	Data protection design in Section 2.8	Design only; legal review not carried out

The proposal states that no deployed system provides Sinhala-language emergency detection. That statement is not repeated here as a finding, because it requires a documented search to support it. The thesis instead claims the narrower and verifiable position that the reviewed sources do not describe such a system.

1.3 Research Problem

Tourists who face an emergency in Sri Lanka have to solve several problems at the same moment: deciding which service to contact, finding the number, explaining what has happened in a language that the responder may not speak, and communicating where they are. At the same time, responders receive alerts that may be incomplete, in another language, without a verified identity or without a precise location, so the first minutes, which matter most, are spent on clarification instead of dispatch. Figure 1.2 shows the main causes and effects.

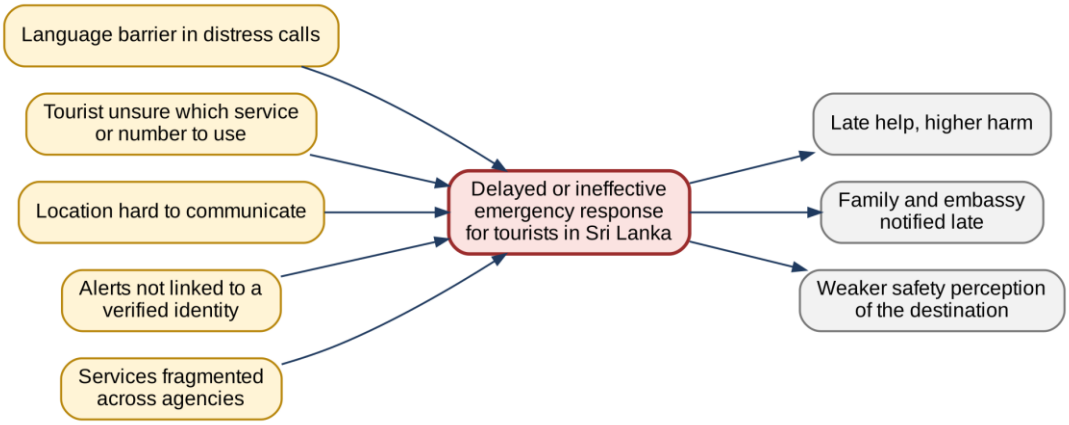


Figure 1.2: Problem tree: causes and effects of delayed emergency response for tourists

The first approach to classifying emergency statements is a static keyword list, and the component keeps such a list as a fallback and as a baseline. Keyword rules are transparent but brittle: they miss paraphrases, confuse incidents that share words and need manual editing whenever a new way of describing an emergency appears. A misclassified incident can be routed to the wrong responder. The research problem is therefore stated as follows:

> **Research problem.** How can a tourist's emergency statement, given by a one-tap manual SOS or by a typed or spoken statement in one of several languages, be classified reliably into an incident type and a responder channel, combined with the tourist's verified profile and GPS location, and delivered to the nearest police station dashboard so that police response is faster and better informed, while the sensitive data involved remain protected?

The following research questions guide the study.

- **RQ1.** Does a Random Forest on TF-IDF features classify held-out tourist emergency statements into incident types better than keyword rules and a majority-class baseline, and how does it compare with logistic regression on the same features?
- **RQ2.** Can a second Random Forest predict the responder channel (police, medical or fire) with usable accuracy?
- **RQ3.** Does a GPS-anchored alert with the tourist’s profile and a structured briefing give responders more information at the start of the response than a voice-only call?

- **RQ4.** Can the complete workflow (one-tap SOS, classification, routing, real-time dashboard, offline queue, shared login) be delivered reliably, so that an alert is never lost when classification or voice processing fails?
- **RQ5.** Is the interface acceptable to its intended users, as judged by stakeholder review now and by a System Usability Scale study in future work?

1.4 Research Objectives

1.4.1 Main objective

To design, implement and evaluate an AI-assisted tourist emergency detection and police response platform for Sri Lanka that combines a one-tap manual SOS, classification of emergency statements, verified tourist profiles, GPS-based routing to the nearest police station and a police-facing dashboard, with explicit protection of personal data.

1.4.2 Specific objectives

1. To analyse the requirements of tourists and police officers for emergency reporting and response, using the literature, the project proposal and stakeholder consultation.
2. To build a labelled dataset of tourist emergency statements with a defined taxonomy of incident types and responder channels, following a documented labelling and quality-control procedure.
3. To train and compare Random Forest and logistic regression classifiers for incident type and Random Forest for responder channel, and to evaluate them against keyword-rule and majority-class baselines.
4. To implement a tourist web application with registration, consent, manual SOS, text or voice statement input and GPS capture.
5. To implement nearest-station routing from stored coordinates using the haversine distance, together with an offline queue so that alerts raised without connectivity are replayed.

6. To implement police and immigration dashboards with real-time updates, structured chat briefings, and acknowledge, escalate and close actions for police operators.
7. To implement an optional voice path (speech-to-text and translation to English before classification), and to document its status and limitations honestly.
8. To design and test security and privacy controls, including token-based authentication, role-based access, duplicate-alert control and consent-based data release.
9. To evaluate the models with hold-out metrics, the platform with automated tests and scenario walkthroughs, and to report measured results separately from planned evaluation.

1.4.3 Mapping of objectives to research questions and evidence

Table 1.4: Mapping of specific objectives to research questions, methods and evidence

Objective	Research question	Main method	Evidence presented in
1 Requirements	All	Literature review, proposal analysis, stakeholder analysis	Sections 2.1, 2.6
2 Dataset	RQ1, RQ2	Taxonomy, labelling, agreement, splitting	Sections 2.3, 3.1.2
3 Model comparison	RQ1, RQ2, RQ3	Stratified hold-out; keyword and dummy baselines	Sections 2.4, 3.1.3
4 Tourist application and routing	RQ3, RQ4	Prototype development, functional tests, scenarios	Sections 2.5, 2.10, 3.1.1
5 Nearest-station routing and offline queue	RQ4	Haversine search, replay tests	Sections 2.5, 2.10
6 Police and immigration dashboards	RQ3, RQ5	Prototype development, stakeholder review	Sections 2.5, 3.1.5
7 Voice and translation path	RQ4	Implementation, failure-path tests	Sections 2.4.7, 3.1.3
8 Security and privacy	RQ4	Design, authorisation and duplicate tests	Sections 2.8, 3.1.6
9 Evaluation	RQ1–RQ5	Hold-out metrics, tests, scenarios (Section 2.10)	Chapter 3

1.4.4 Scope, delimitations and organisation of the thesis

The study is limited to the SOS component of TourCeylon. It covers text classification of emergency statements, the tourist web application, the police and immigration dashboards, nearest-station routing, the offline queue and data protection design. It does not cover connection to the 119 or 1912 call centres, live ambulance tracking, medical diagnosis, integration with immigration databases or a native mobile application, and it was not trained or tested on real victims' statements. Statements about integration with the Police Department are limited to the proposal submitted and to the permissions that are documented in Appendix E. Chapter 2 describes the methodology, the architecture, the requirements, the social, security and ethical aspects, the commercialisation aspects and the testing and implementation. Chapter 3 presents the results and discussion, Chapter 4 concludes, and the appendices contain the instruments, test cases and supporting material.

2. METHODOLOGY

This chapter describes how the SOS Emergency Safety System was designed, how it was built and how it was evaluated. The status of each element is stated explicitly. Implemented means that the author built it and it runs in the current system. Proposed means that it is described in the project proposal or design but is not claimed as working. Planned means that it is evaluation or development work still to be done, and measured means that a result exists in the evidence record. Chapter 3 reports the measured results, which are limited to a hold-out comparison of classifiers, automated tests and scenario walkthroughs. Items that the author must still complete are highlighted.

2.1 Methodology

2.1.1 Research approach

The research follows design science [51], in which knowledge is produced by building and evaluating an artefact that addresses a relevant problem. The process model of Peffers et al. [52] was adopted: problem identification (tourist emergency communication), definition of objectives, design and development of the artefact (the SOS component), demonstration through scenarios, evaluation through classifier metrics and automated tests, and communication through this thesis (Figure 2.1).

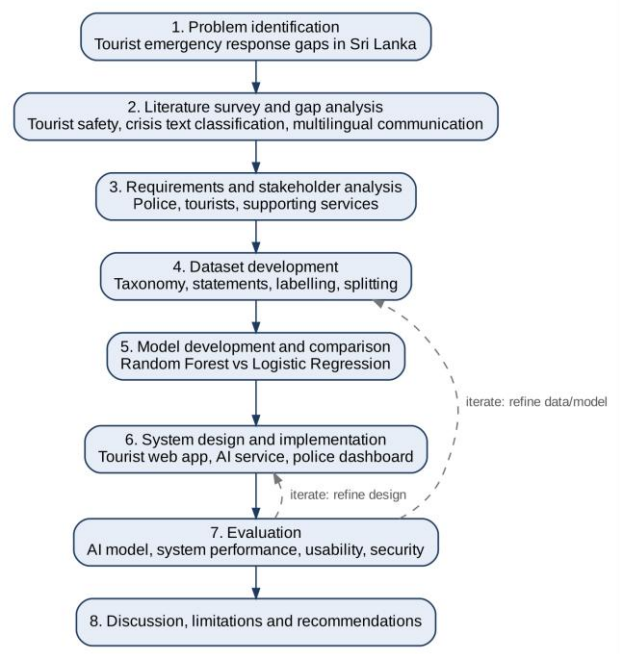


Figure 2.1: Overall research methodology of the SOS component

Design science suits the work because the value of an AI-assisted, map-backed SOS can only be assessed once such a system exists. Development was iterative (Figure 2.2) [53]. Two examples show how evaluation changed the design. Early versions displayed raw tourist statements to police, including filler words and repeated phrases from voice transcripts, which led to the statement-cleaning step. Early hold-out results on the fourteen-class task led to the comparison of several candidate models instead of reliance on a single algorithm.

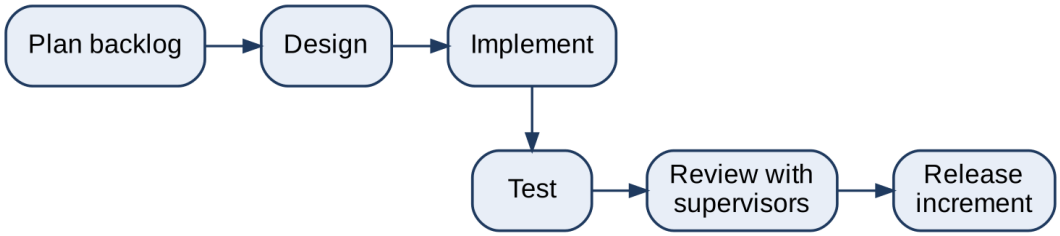


Figure 2.2: Iterative development lifecycle used in the project

2.1.2 Stakeholders and requirements elicitation

Four groups of stakeholders influence the design (Figure 2.3). Tourists need a way to raise an alarm quickly. Police and Tourist Police officers need an alert that says who the tourist is, what has happened and where. Medical, fire and immigration officers need the information appropriate to their role. Tourism authorities, hotels and tour

operators need aggregated, privacy-respecting insight and a duty-of-care tool for guests.

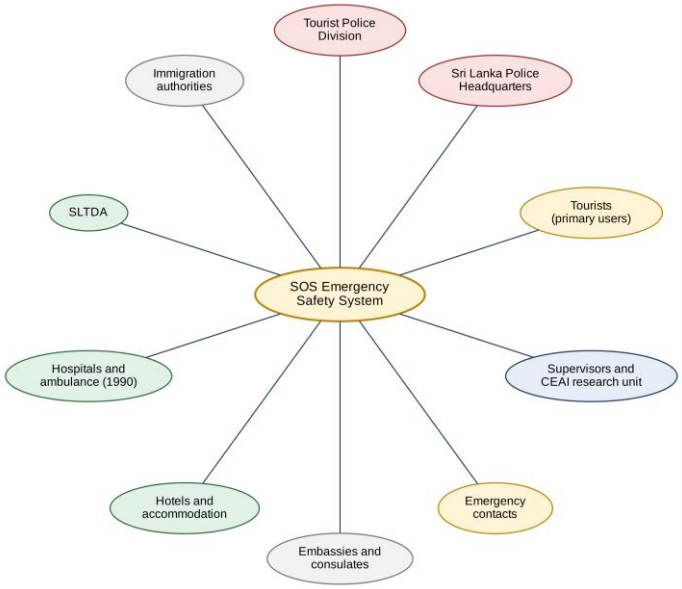


Figure 2.3: Stakeholders of the SOS system and their interests

Requirements were elicited from four sources: the project proposal submitted to the Sri Lanka Police, the literature and public reports reviewed in Chapter 1, review by the supervisors and stakeholders of the screens and police briefing formats, and informal conversations with tourists. The author reports that the conversations took place in Colombo, Galle Face, Port City Colombo and Negombo and involved approximately 100 people [confirm number, profile and purpose of these conversations]. They are treated only as qualitative input to requirements. They did not produce training data, because the corpus is written by the author (Section 2.3), and they are not treated as a statistical sample.

2.2 Application of Key Pillars of the Specialisation

The Information Technology specialisation expects the final product to apply the key knowledge areas of the field systematically. Table 2.1 lists the pillars applied in this research and where each appears in the thesis.

Table 2.1: Knowledge pillars of the specialisation applied in the component

Pillar	How it is applied	Section
Requirements engineering	Stakeholder analysis, functional and non-functional requirements, use cases and role permissions	2.1, 2.6
Software architecture and design	Layered architecture, deployment view, sequence, lifecycle and data model, design alternatives	2.5, 2.7
Machine learning and NLP	Supervised text classification, TF-IDF features, Random Forest versus Logistic Regression, baselines, class-imbalance aware evaluation	2.3, 2.4
Data engineering	Taxonomy, corpus guidelines, de-identified synthetic data, stratified leakage-free splitting	2.3
Database systems	Relational storage of tourists, incidents, stations, chat and journey points	2.5
Web and real-time engineering	React clients, REST API, WebSocket events, offline queue	2.5
Information security and privacy	JWT authentication, role-based access, consent, duplicate control, Personal Data Protection Act	2.8
Software quality assurance	Automated tests, scenario walkthroughs, stakeholder review	2.10
Project management and entrepreneurship	Iterative increments, group integration, pilot plan and business model	2.1, 2.9

2.2.1 Evaluation metrics

Because the system must not miss genuine emergencies, evaluation does not rely on accuracy alone. For a class c, precision, recall and the F1-score are computed from true positives (TP), false positives (FP) and false negatives (FN) [35]:

```
precision_c = TP_c / (TP_c + FP_c)
recall_c    = TP_c / (TP_c + FN_c)
F1_c        = 2 * precision_c * recall_c / (precision_c + recall_c)
accuracy    = (number of correct predictions) / N
macro-F1    = (1 / K) * sum over classes c of F1_c        (K = number of
classes)
```

Macro-averaged F1 gives each incident type equal weight and therefore exposes weak performance on rare classes, which accuracy can hide [34]. This matters for emergency triage, where rare incidents such as fire count as much as frequent ones such as theft. The macro-F1 of a dummy classifier that always predicts the most frequent class and of a keyword-rule classifier give reference points that a useful model must beat.

2.2.2 Nearest-station computation

The nearest station is found with the haversine great-circle distance between the incident's latitude and longitude and each stored station of the required type. With latitudes φ and longitudes λ in radians and Earth radius R ≈ 6371 km [43]:

```
a = sin^2((phi2 - phi1)/2) + cos(phi1) * cos(phi2) * sin^2((lambda2 -
lambda1)/2)
d = 2 * R * asin( sqrt(a) )
```

The station with the smallest d among those of the type chosen by the responder classifier is selected. The result is a straight-line distance and not a road distance. Travel time could differ substantially, particularly in congested areas, and this limitation is discussed in Section 3.2.

2.3 Dataset Methodology

2.3.1 Corpus and taxonomy

The principal study material is a corpus of 264 labelled statements covering 14 incident types, held in the file sos_ml_corpus.py. Each record contains a tourist-style statement, an incident type and a responder label (police, medical or fire). The incident types are aligned with the incident types stored by the application (Incident.VALID_TYPES), so that the labels predicted by the model are exactly those the dashboards display and the analytics use. Table 2.2 gives representative statements. The complete list of the 14 types and the number of statements per type is in Appendix A [insert list from Incident.VALID_TYPES and counts per class].

Table 2.2: Illustrative statement styles in the corpus (paraphrased examples)

Statement style	Incident type	Responder
someone grabbed my bag near the station and ran	Theft	Police
my friend collapsed, not breathing properly	Medical	Medical
a man keeps following us and touching my friend	Harassment	Police
there is smoke and fire in the guesthouse kitchen	Fire	Fire
our tuk tuk crashed, driver is bleeding	Road accident	Medical
we are lost on the trail and it is getting dark	Unsafe location / lost	Police

Source: Examples from the author's corpus.

2.3.2 Source of the data and corpus guidelines

All statements were written or paraphrased by the author to resemble what a tourist might type or say in an emergency: short, informal, sometimes ungrammatical and sometimes mixing several facts. No real incident reports and no real victims' words were used, for ethical reasons. This is an important limitation: the corpus is synthetic, so accuracy on it may not carry over to real speech (Section 3.2.4). To make the corpus as realistic as possible, explicit guidelines were followed. Each incident type includes direct statements (my wallet was stolen), indirect descriptions (my bag is gone from the seat), statements written in a panicked style with missing or repeated words, statements that mention landmarks, and statements as they might appear after machine translation from another language. Each statement also carries a responder label so that the responder model learns from the same text. Statements that genuinely span several incident types were labelled with the most operationally urgent type, for example injury over theft when someone is hurt during a robbery.

2.3.3 Labelling, privacy and ethics

Because the author wrote and labelled the statements, there is a single annotator and inter-annotator agreement (Cohen's kappa) could not be computed [54]. This is reported as a limitation. No personal data are contained in the corpus. No human participants were recruited to provide data. Further study material comprises the seeded police and medical station tables (seed_sos_data.py) used for routing and synthetic demonstration tourist profiles used for testing.

2.3.4 Cleaning and splitting

Each statement is cleaned (lower-casing, whitespace normalisation, removal of filler words and duplicated phrases). The corpus is split into a stratified training set (80%) and a hold-out test set (20%) with random_state = 42, so that every class is represented in both where possible. This gives about 53 test statements. The TF-IDF vectoriser is fitted on the training portion only, which prevents vocabulary from the test set from leaking into training [36]. After evaluation, the production models are refitted on all 264 statements and saved. Table 2.3 is the dataset record.

Table 2.3: Dataset description record

Item	Value
Labelled statements	264
Incident types / responder categories	14 / 3 (police, medical, fire)
Source	Written or paraphrased by the author; synthetic; no real incident data
Languages	English; non-English input is translated to English before classification
Train / test split	Stratified 80/20, random_state = 42; about 53 test statements
Class distribution and imbalance	[insert counts per class (Appendix A)]
Inter-annotator agreement	Not applicable (single annotator)
Duplicates removed	[TBD: confirm whether duplicate statements were removed before splitting]

2.4 AI Model Development and Experimental Design

2.4.1 Classification pipeline

The workflow is: emergency statement, cleaning, TF-IDF features, Random Forest incident-type classification, Random Forest responder classification, urgency mapping, nearest-station lookup and dispatch (Figure 2.4). The first Random Forest identifies what has happened. The second decides which service should respond. A keyword classifier serves as the runtime fallback if the model bundle cannot be loaded, so that an alert is never blocked by the AI component.

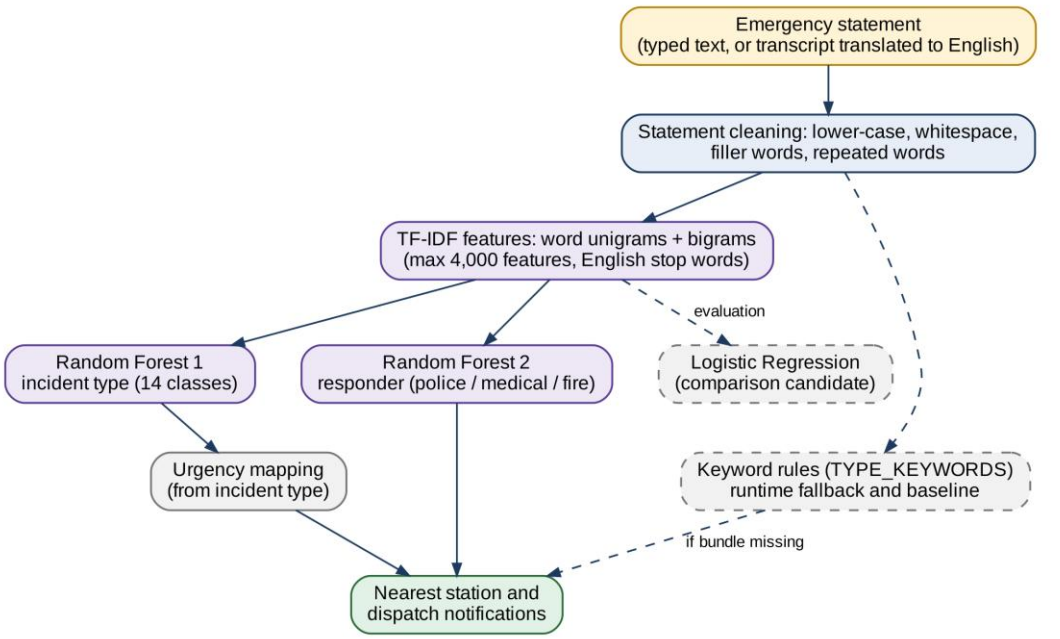


Figure 2.4: Text classification pipeline of the SOS system

2.4.2 Preprocessing and features

Text is converted into numerical features with TF-IDF weighting [23], [55] over word unigrams and bigrams, which preserves short phrases such as “chest pain” and “bag stolen”. The vectoriser uses at most 4,000 features and English stop words. TF-IDF is fast, transparent and needs no large pre-trained model, which suits a lightweight server deployment. Non-English input (for example a Sinhala or Tamil voice message) is translated into English before classification, so the vectoriser and the classifiers are English-centric (Section 3.2.4).

2.4.3 Classifiers and configuration

Random Forest [26] builds many decision trees on bootstrap samples with random feature subsets and takes a majority vote. It handles sparse high-dimensional features and provides probability estimates that are shown to operators as a confidence score. It was also chosen to remain consistent with the Random Forest modules used elsewhere in the TourCeylon group platform. Logistic Regression [24] is a linear model that is the standard baseline for text. Both are implemented in scikit-learn [28]. Table 2.4 gives the configuration, and simplified code excerpts are in Appendix C.

Table 2.4: Preprocessing steps and model configuration

Stage	Setting
Statement cleaning	Lower-casing, whitespace normalisation, removal of filler words and duplicated phrases
Translation	Non-English input translated to English before classification where available
Vectoriser	TF-IDF, word unigrams and bigrams, max_features = 4000, English stop words; fitted on the training portion only
Incident Random Forest	180 trees; balanced class weights; random_state = 42; 14 classes
Responder Random Forest	Same features and settings; 3 classes (police, medical, fire)
Logistic Regression candidate	Balanced class weights; same TF-IDF features
Baselines	Keyword-rule matcher (TYPE_KEYWORDS); DummyClassifier (most frequent class)
Split	Stratified 80/20, random_state = 42; production refit on all 264 statements

Balanced class weights increase the influence of rare classes during tree construction and partly compensate for imbalance. The number of trees was chosen as a compromise between stability of predictions and model size; beyond this point additional trees produced little change in the hold-out metrics while increasing the size of the saved bundle [confirm this statement of the author's tuning].

2.4.4 Baselines

Two baselines are evaluated on exactly the same hold-out statements. The dummy classifier predicts the most frequent class and shows what could be achieved without reading the text. The keyword classifier uses a hand-written list of indicative words per incident type; it is also the runtime fallback. A learned model is useful only if it improves on these references. This design ensured that the weakness of the Random Forest on the fourteen-class task became visible (Section 3.1.3).

2.4.5 Handling imbalance

With 264 statements across 14 classes, some classes have only a handful of examples. Class weighting is used in both learned models. Resampling methods such as SMOTE [56] are not used in the current system and are listed as a possible extension, because they must be applied only to training data to avoid leakage.

2.4.6 Evaluation protocol

The implemented protocol is a single stratified hold-out evaluation. For each model the training script records accuracy and macro-F1 on the hold-out set, together with per-class precision and recall, and writes them to metrics.json. Misclassifications are inspected per class to separate errors caused by genuinely ambiguous statements from errors caused by sparse training support. Because the hold-out set has only about 53 statements, a single misclassification changes per-class metrics substantially, so the results are interpreted as indicative and not definitive.

The analysis also considers which errors matter most. Confusing two police-handled types, such as theft and robbery, has little operational consequence because the same responder is notified. Confusing a medical case with a police case is more serious, because the wrong service may be dispatched first. For this reason the responder model is evaluated separately.

Three additional analyses are planned and are not claimed as completed: stratified five-fold cross-validation on the training portion [36] to show variability, McNemar's test [37] on paired predictions to test whether the difference between Random Forest and Logistic Regression is statistically significant [38], and a larger corpus. The script run_experiment.py supplied with this thesis performs the first two on the author's corpus.

2.4.7 Voice and translation path

Voice is the most natural channel in an emergency. When the tourist records a voice message, the module chat_voice.py attempts speech-to-text, optionally with Whisper [40], and, where needed, translation into English (translate_service.py) before classification (Figure 2.5). If transcription fails, the audio is preserved and attached to the incident so that the responder can listen to it, and the incident is still created with the GPS position and a fallback label. A voice problem can therefore never block an alert. The project proposal names Google Cloud Translation [42] as the translation service; the provider actually used by translate_service.py and the languages that were tested are [TBD: state provider and tested languages]. The project proposal sets a target of ten languages including Sinhala and Tamil; this is a design target and not a validated

result, because no per-language evaluation has been carried out. Speech recognition quality, if measured, would be reported as word error rate [41]:

```
WER = (S + D + I) / N
S = substitutions, D = deletions, I = insertions, N = words in the
reference transcript
```

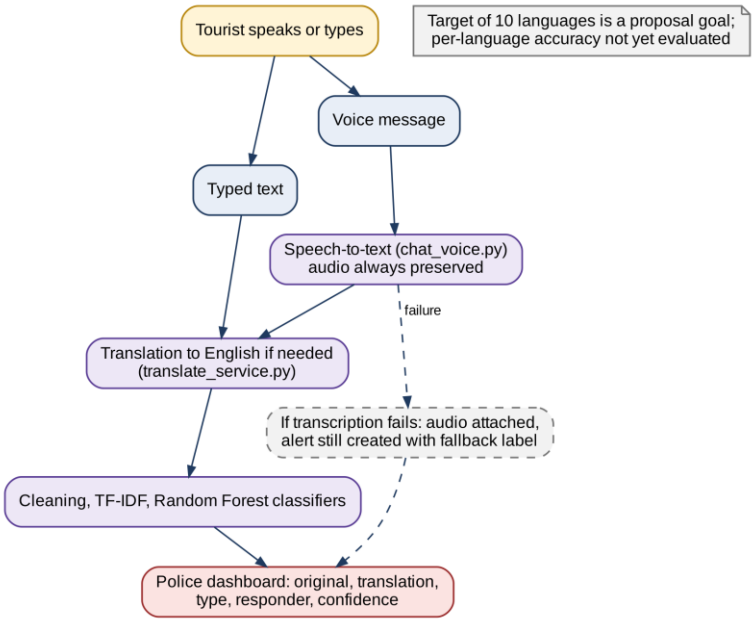


Figure 2.5: Voice and multilingual processing path

The word error rate requires reference transcripts that are written by hand, and none have been produced, so it is reported as not evaluated.

2.5 System Architecture and Design

2.5.1 High-level architecture

The module uses a layered architecture (Figure 2.6). The presentation layer contains three React clients: the tourist client, the police dashboard and a read-only immigration view. The intelligence and application layer is a unified Flask application on port 5002 that exposes a REST API, a Flask-SocketIO channel for real-time events [57], the classification service (TF-IDF with two Random Forests, with keyword fallback), urgency mapping, routing and notification. The data layer is a PostgreSQL database shared with the other TourCeylon components, and seeded station data. External services are the browser Geolocation API, speech-to-text and translation services and the SMS and e-mail provider. Because the application shares the database and the

JSON Web Token login with the other components, an SOS can reference the same tourist identity and itinerary that the planning component uses.

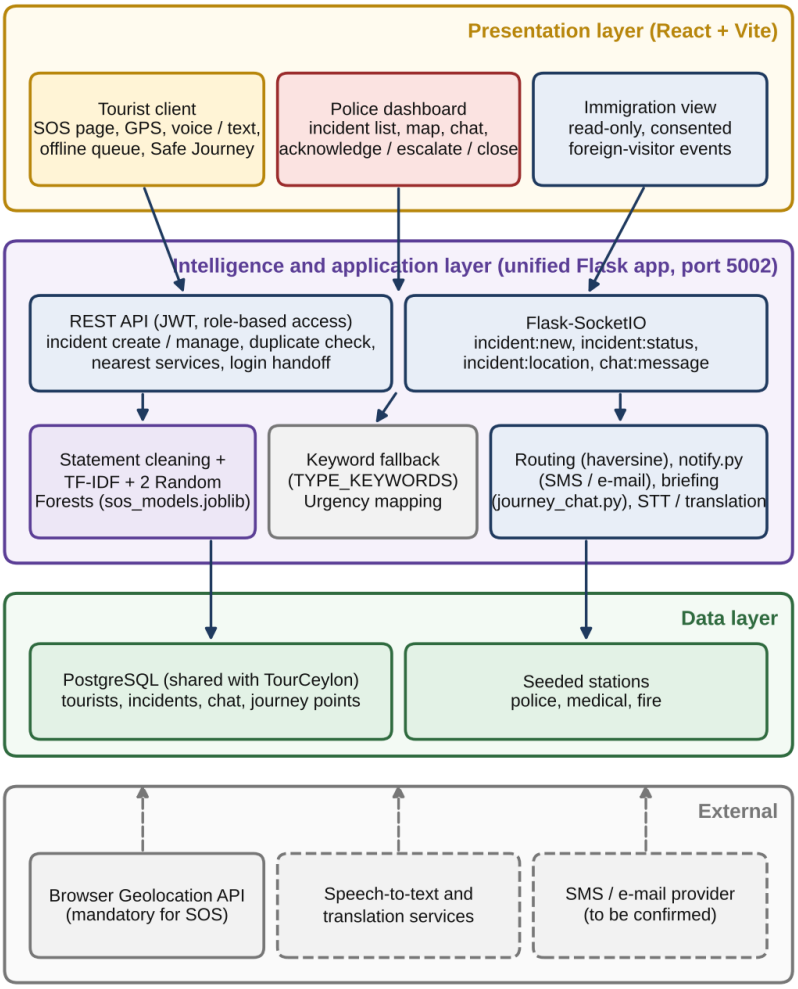


Figure 2.6: High-level architecture of the SOS Emergency Safety System (dashed = optional or external service)

Table 2.5 separates what is implemented from what is proposed. The classification of status follows the author's component notes and the delivered feature list.

Table 2.5: Implementation status of the main components

Component	Status	Notes
Tourist web client	Implemented	One-tap SOS, mandatory GPS, text and voice, offline queue, Safe Journey; no native mobile app
Police dashboard	Implemented	Real-time list and map, chat, acknowledge, escalate, close
Immigration view	Implemented	Read-only, consented foreign-visitor events

Component	Status	Notes
Incident classification	Implemented	Two Random Forests; Logistic Regression is a comparison candidate; keyword fallback
Nearest-station routing	Implemented	Haversine distance over seeded stations
Notifications and briefing	Implemented	SocketIO, SMS and e-mail templates, structured chat briefing
Offline queue	Implemented	Local queue and replay with server-side duplicate check
Voice path	Implemented (optional)	Speech-to-text and translation; audio preserved on failure; per-language accuracy not evaluated
Login handoff from TourCeylon	Implemented	Token exchange so SOS needs no second login
Named-entity recognition	Proposed	Described in the proposal; not part of the evaluated pipeline
Ten-language support	Proposed target	Not validated
Integration with Police operational systems	Not claimed	Phased plan only
Connection to immigration databases	Not claimed	The immigration view shows consented SOS events only

2.5.2 Deployment view

Figure 2.7 shows the deployment. The tourist, police and immigration clients are served as web applications and call the unified back end over HTTPS. For development the tourist client runs on a dedicated Vite port with a proxy to the API. The back end loads the serialised model bundle (sos_models.joblib) once and keeps it in memory. Because inference uses a small bundle, retraining and redeployment take only seconds, which allows the corpus to be extended frequently. To run the module: start the unified back end (port 5002), run seed_sos_data.py to load the stations, run RETRAIN_SOS_ML.bat after corpus updates, and start the client. Hosting details for a production deployment (provider, region, TLS configuration) are [TBD].

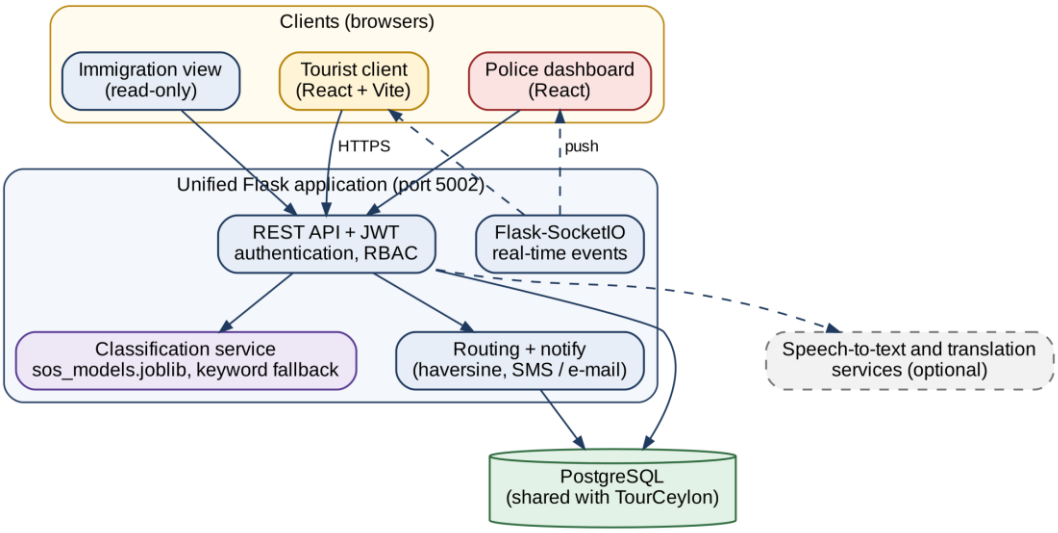


Figure 2.7: Deployment view of the web-based system

2.5.3 Actors, use cases and role permissions

The actors are the tourist, the police and responder operator, the immigration officer and the administrator (Figure 2.8). Table 2.6 lists the principal use cases and Table 2.7 the permissions of each role, which are verified by the authorisation-matrix test (Section 2.10).

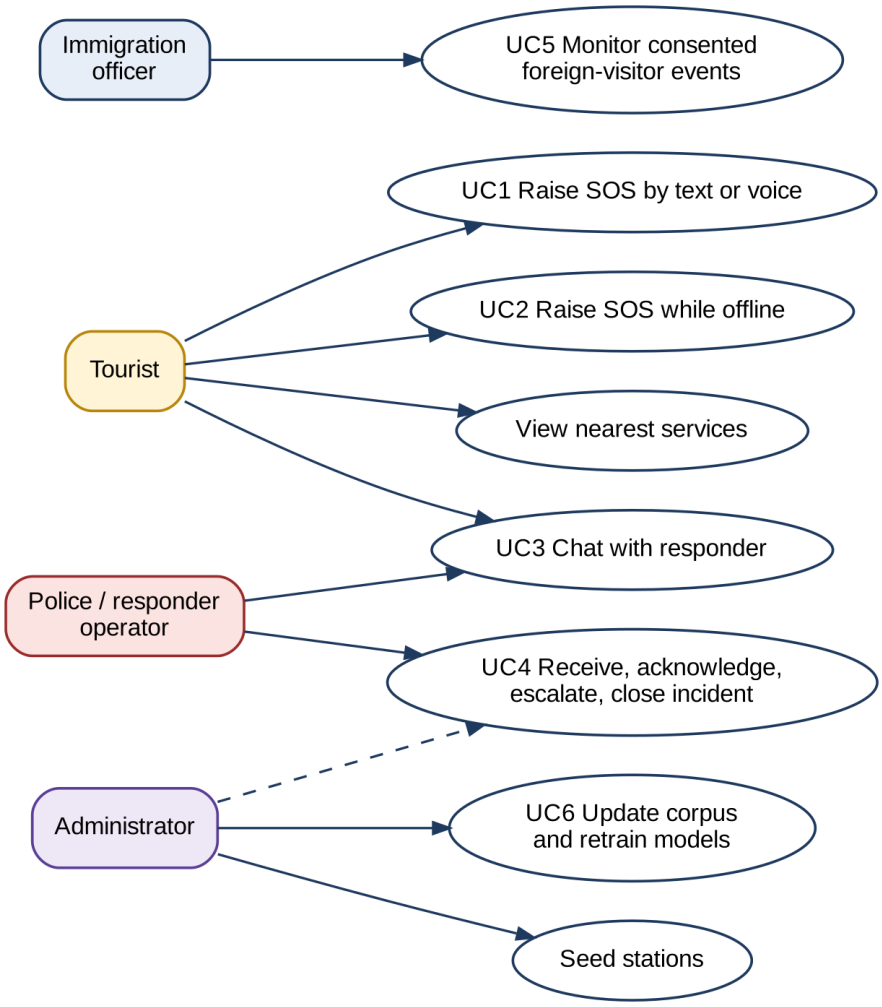


Figure 2.8: Use case diagram of the SOS system

Table 2.6: Principal use cases

ID	Actor	Use case	Outcome
UC1	Tourist	Raise SOS by text or voice	Incident created with GPS, type, urgency and responder
UC2	Tourist	Raise SOS while offline	Alert queued and replayed on reconnection
UC3	Tourist	Chat with responder	Two-way messages linked to the incident
UC4	Police operator	Receive, acknowledge, escalate, close	Incident status updated; tourist informed
UC5	Immigration officer	Monitor consented foreign-visitor events	Read-only oversight view
UC6	Administrator	Update corpus and retrain	New sos_models.joblib and metrics.json

Table 2.7: Role-based permissions

Action	Tourist	Police	Immigration	Admin
Create SOS and view own incidents	Yes	No	No	No
View station incidents and map	No	Yes	No	Yes
Acknowledge, escalate and close	No	Yes	No	Yes
Chat on incident	Own	Assigned	No	Yes
Monitor consented foreign-visitor events	No	No	Read-only	Yes
Seed stations and retrain models	No	No	No	Yes

2.5.4 SOS workflow

Figure 2.9 shows the flow of an SOS event. When a tourist presses the SOS button, the client captures the statement and the location. If no GPS position is available, the request is rejected with guidance on enabling location, and the hotline numbers 119 and 1912 are displayed. Otherwise the server checks for duplicates, cleans and classifies the statement, finds the nearest station of the responder type, stores the incident, emits a SocketIO event to the station and sends SMS and e-mail notifications. It also posts a structured briefing into the incident chat. If classification fails, the alert is still delivered with a fallback label. This separation between guaranteed steps (GPS capture, profile attachment, notification) and estimated steps (type and responder) is a design principle: a classification error degrades the briefing but never prevents an alert from being delivered.

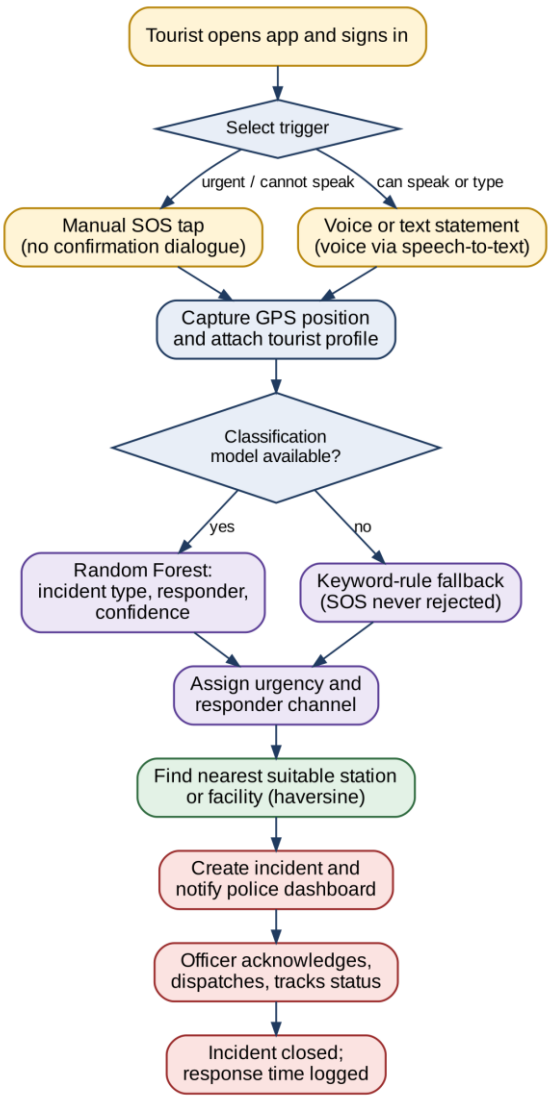


Figure 2.9: Flow of an SOS event from the tourist to the police dashboard

The order of the interactions is shown in Figure 2.10.

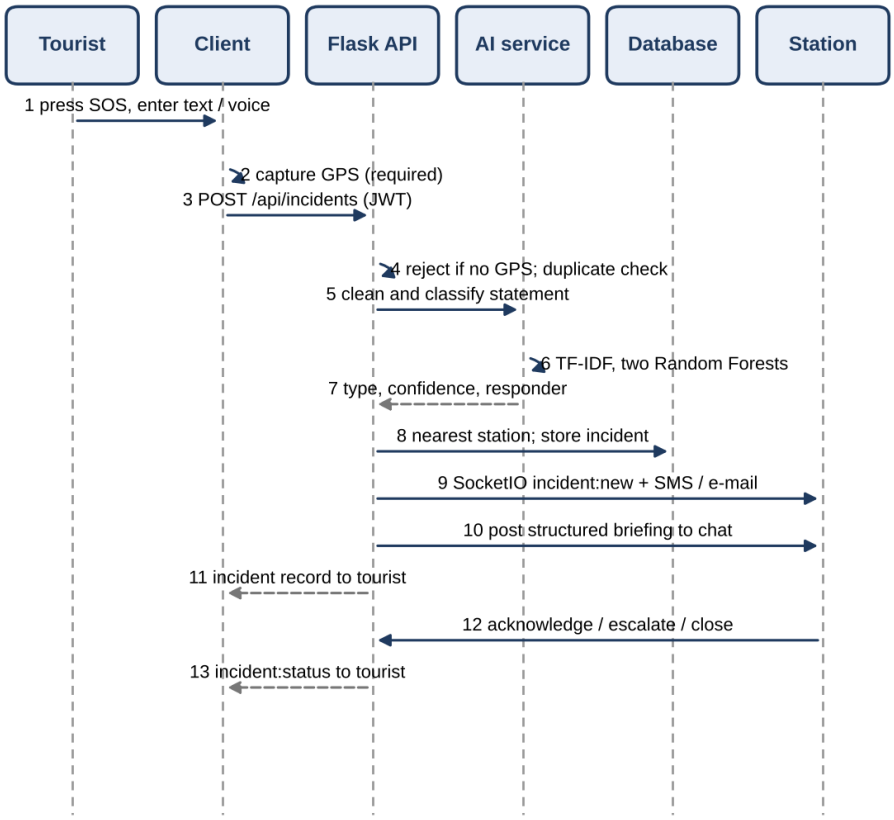


Figure 2.10: Sequence diagram of incident creation

2.5.5 Routing and urgency mapping

Each incident type is mapped to an urgency level (Table 2.8). Medical collapse and fire are priority-one events, which trigger immediate dispatch notifications. The responder model's output determines the type of station, and the routing module computes the haversine distance from the incident coordinates to every seeded station of that type and selects the nearest (Figure 2.11). If the GPS accuracy is poor, the accuracy radius is shown to the officer.

Table 2.8: Urgency mapping (illustrative)

Incident type	Urgency	Default responder
Medical (collapse, breathing, serious injury)	Priority 1	Medical
Fire	Priority 1	Fire
Road accident with injury	Priority 1	Medical / Police
Assault or harassment in progress	Priority 1–2	Police
Theft or robbery	Priority 2	Police
Lost or unsafe location	Priority 2–3	Police

Incident type	Urgency	Default responder
Remaining types	[insert from the urgency map in the code]	[insert]

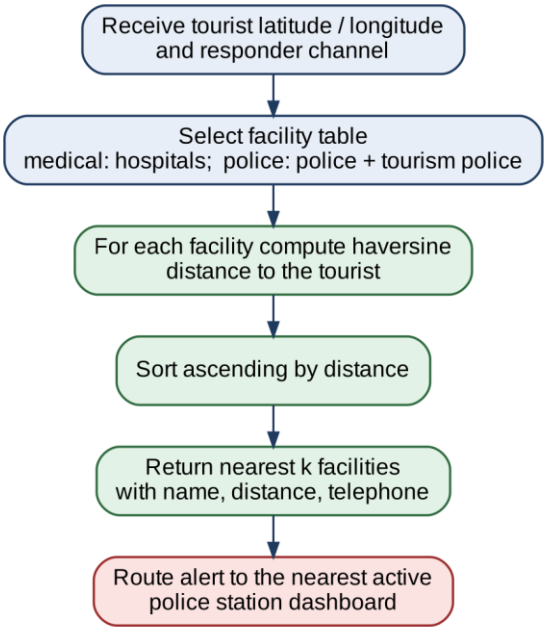


Figure 2.11: Nearest-station selection logic

2.5.6 Incident lifecycle and real-time events

An incident is created with GPS and statement, classified and routed, acknowledged by a station and eventually closed, with an optional escalation step (Figure 2.12). Each transition is broadcast over SocketIO so that the tourist and all relevant dashboard users see the current status. Table 2.9 and Table 2.10 list representative interfaces.

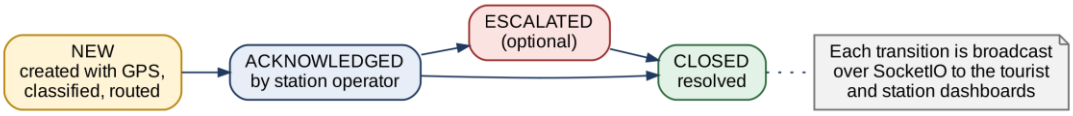


Figure 2.12: Incident lifecycle

Table 2.9: Principal REST endpoints (representative paths)

Method and path	Purpose
POST /api/incidents	Create SOS with GPS, statement and optional journey points
GET /api/incidents/mine	Tourist's own incidents and statuses
GET /api/police/incidents	Incidents for the operator's station

Method and path	Purpose
POST /api/police/incidents/{id}/ack | /escalate | /close	Lifecycle actions
POST /api/incidents/{id}/chat	Send chat message (text or voice)
POST /api/incidents/{id}/location	Live GPS update
GET /api/services/nearby	Nearest police and medical services
POST /api/tourists/token-exchange	TourCeylon login handoff

Source: Representative paths from the author's design; [confirm names against the code].

Table 2.10: SocketIO events (representative names)

Event	Direction	Payload
incident:new	Server to station room	Summary with type, urgency, confidence and map link
incident:status	Server to tourist and station	New status after acknowledge, escalate or close
incident:location	Server to station room	Latest coordinates and accuracy
chat:message	Both directions	Text, sender, time, optional translation

2.5.7 Briefing and notifications

The police chat briefing produced by journey_chat.py consolidates in one message the predicted incident type with confidence, the urgency, the cleaned statement, the tourist's name and nationality, contact details, a map link to the current position and, where available, a link to recent Safe Journey route history. The most actionable information (type, urgency and location) comes first and a final line reminds the operator that the AI label is a suggestion and the original statement is shown. This ordering was refined after review, when early versions that listed profile details first were found to slow down reading. Short SMS and e-mail notifications (notify.py) link to the dashboard because SMS may be read on a basic phone.

2.5.8 Offline queue and duplicate control

If the network request fails, the client stores the payload in the browser's local storage and replays it when the browser reports that it is online again (Figure 2.13). The server also handles replayed requests, and a duplicate check ignores a second alert from the same tourist within a short interval, so that a request that had in fact succeeded is not delivered twice.

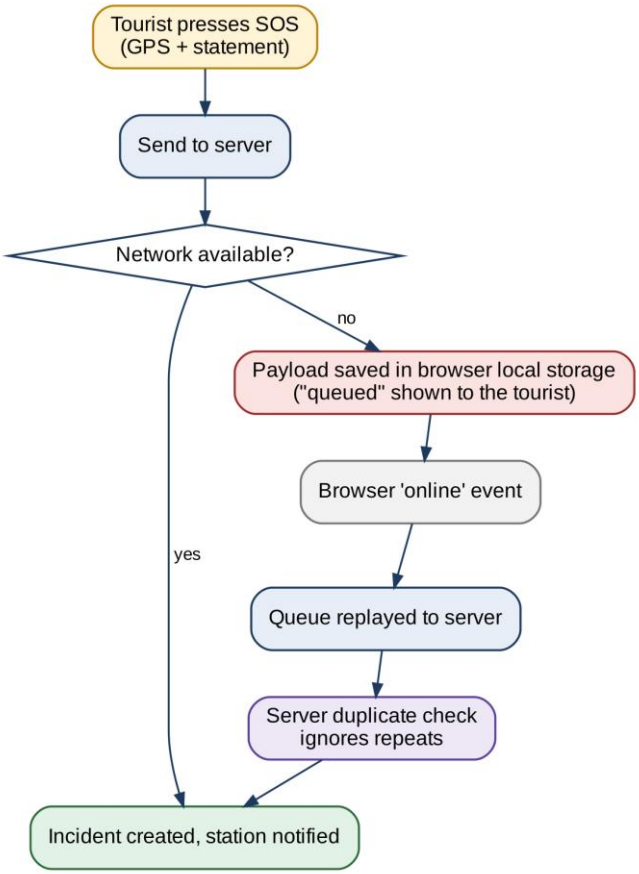


Figure 2.13: Offline queuing and replay of SOS alerts

2.5.9 Data model

Figure 2.14 gives the conceptual data model. A tourist has zero or more incidents and journey points. An incident is assigned to a station and has chat messages. The figure is conceptual; the authoritative schema is defined in the SQLAlchemy models.

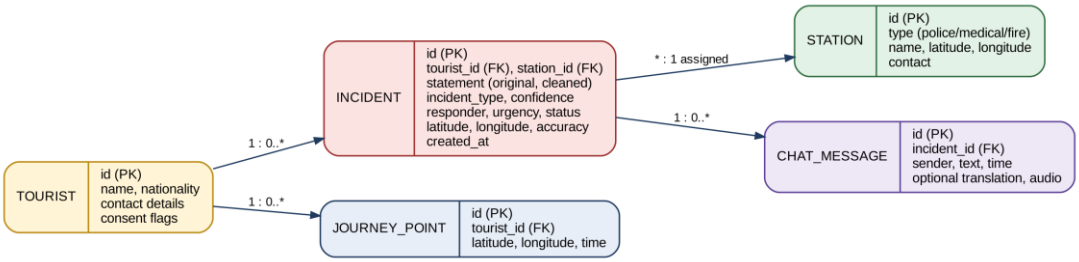


Figure 2.14: Conceptual entity-relationship diagram

2.5.10 External interfaces

Table 2.11: External interfaces of the system

Interface	Direction	Data exchanged	Status
Browser Geolocation API	Device to client	Latitude, longitude, accuracy	Implemented; permission required
Speech-to-text	Back end to model or service	Audio, transcript	Implemented (optional path)
Translation	Back end to service	Transcript or chat text	Implemented; [TBD provider]
SMS and e-mail	Back end to provider	Short alert with dashboard link	Implemented; [TBD provider]
TourCeylon login	Itinerary client to SOS	Access token exchange	Implemented
119 and 1912 hotlines	Display only	Numbers shown on every SOS screen	Implemented; no technical integration
Police operational systems	Bidirectional	Alert, status	Not implemented; future work

2.6 Requirements

2.6.1 Functional requirements

Table 2.12: Functional requirements

ID	Requirement	Priority
FR1	One-tap SOS for authenticated tourists, with mandatory GPS capture	Must
FR2	Accept the incident description as text or voice (speech-to-text with optional translation)	Must
FR3	Label the incident type with AI and show a confidence score; fall back to keyword rules if models are unavailable	Must
FR4	Determine the responder channel (police, medical or fire) and route to the nearest suitable station	Must
FR5	Stream live GPS updates and attach Safe Journey route history when enabled	Should
FR6	Provide two-way chat between tourist and responder, with a structured briefing message	Must
FR7	Queue SOS alerts offline and replay them when connectivity returns	Must
FR8	Show nearest services (police, hospitals) to the tourist	Should
FR9	Allow police to acknowledge, escalate and close incidents	Must
FR10	Notify stations through real-time dashboard updates and SMS or e-mail	Must
FR11	Provide a read-only view of consented foreign-visitor events to immigration officers	Should

2.6.2 Non-functional requirements

The non-functional requirements follow the quality characteristics of ISO/IEC 25010 [58] (Table 2.13). Targets are design targets and are not measured results unless Chapter 3 states otherwise.

Table 2.13: Non-functional requirements

ID	Category	Requirement
NFR1	Performance	Sub-second classification on typical statements
NFR2	Latency	Real-time SocketIO delivery suitable for operational demonstrations; the proposal target is under 90 seconds from detection to dispatch
NFR3	Usability	Mobile-friendly interface usable under stress, with large controls and minimal steps
NFR4	Reliability	Alerts are delivered even if classification fails (fallback) or the network is down (offline queue)
NFR5	Security	Role-based access for tourist, police and immigration roles; JWT authentication
NFR6	Privacy	Location and profile shared only with consent at SOS time
NFR7	Maintainability	Models retrained by script without code changes; classifier replaceable

2.6.3 Interface design for users under stress

The tourist interface was designed for people who may be frightened, injured or in a hurry. The SOS control is large and placed where it can be reached with one thumb, the statement field accepts very short input, the voice option requires a single press, and status messages use plain language such as “Help request sent to nearest police station”. The official numbers 119 and 1912 remain visible on every SOS screen so that the tourist can always call if preferred. The AI label is shown with a short explanation that it is an automatic suggestion.

2.7 Technology Selection and Design Alternatives

Table 2.14: Technology stack and justification

Layer	Technology	Justification
Back end	Python 3, Flask, Flask-SocketIO, SQLAlchemy	Same language as the machine-learning code, so classification runs in-process; real-time events fit the Flask stack of the group [57]
Machine learning	scikit-learn, joblib [28]	TF-IDF, Random Forest, Logistic Regression, baselines and serialisation in one library

Layer	Technology	Justification
Front end	React with Vite	Component reuse, responsive layout, common stack across the TourCeylon clients
Database	PostgreSQL	Relational integrity and a single store shared by the group components
Location	Browser Geolocation API	No installation; permission-based; reports accuracy
Voice	Optional Whisper or other STT [40]	Multilingual transcription; failure never blocks an alert
Authentication	JSON Web Tokens [47]	Stateless sessions and a single TourCeylon login

Table 2.15 records the alternatives that were considered and why they were not selected.

Table 2.15: Design alternatives considered

Alternative	Advantages	Reason not selected
Deep-learning transformers [29], [30]	Strong language understanding	Small corpus; heavier deployment; harder to explain
Manual operator-only triage	Human judgement on every case	Does not scale; slow when operators are busy; no structured data
Keyword rules only	Simple and predictable	Fails on paraphrase; retained only as fallback and baseline
Standalone SOS application	Independent release cycle	No shared identity or itinerary context with TourCeylon; tourists rarely install an app just in case
TF-IDF + Random Forest with keyword fallback (selected)	Lightweight, explainable, consistent with group stack, robust when models are missing	Lower fourteen-class macro-F1 than Logistic Regression on sparse data (Section 3.1.3)

Why a web application. A browser-based module can be used by tourists without installing software and allows rapid iteration. Its limits are reduced access to background sensors and push notifications compared with a native application, which is future work.

2.8 Social, Security and Ethical Aspects and Limitations

2.8.1 Security design

Security is designed in layers (Figure 2.15). All SOS endpoints require a valid JSON Web Token [47], and role claims determine which endpoints and records a user can access. An incident is visible only to the reporting tourist, the assigned station and administrators. Location updates are accepted only for open incidents owned by the

tourist. A duplicate check limits accidental or malicious repeated alerts. These measures are verified by the authorisation-matrix and duplicate tests (Section 2.10). The design follows REST conventions and the OWASP Top Ten [49] as a reference, and ISO/IEC 27001 [50] is used only as a reference; the thesis does not claim certification. Multi-factor authentication [46], a full audit log, encryption at rest, a retention schedule and TLS configuration in production are not claimed as implemented and are listed as planned improvements.

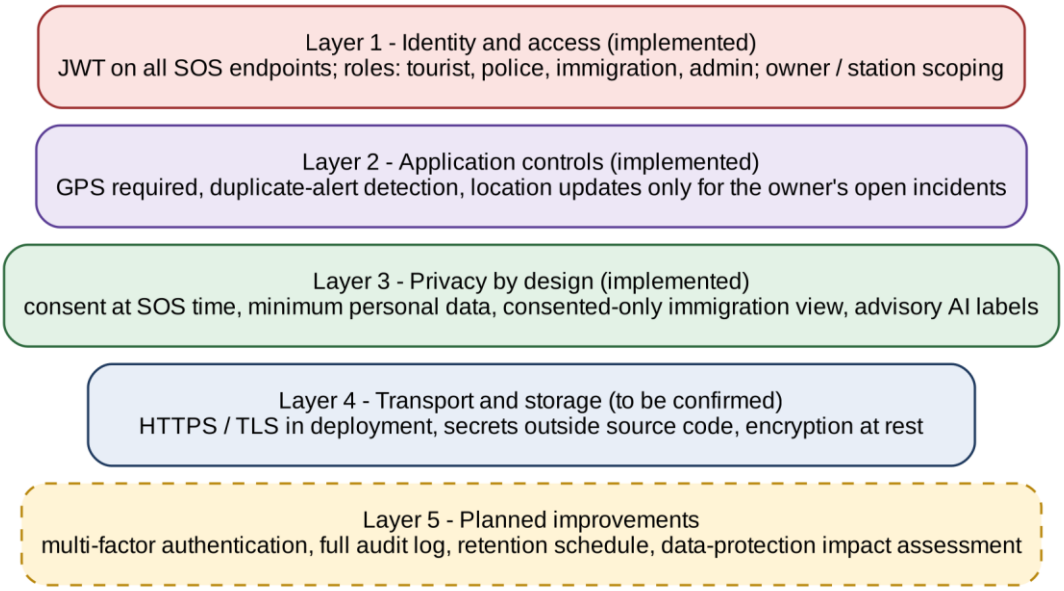


Figure 2.15: Layered security design (implemented and planned layers)

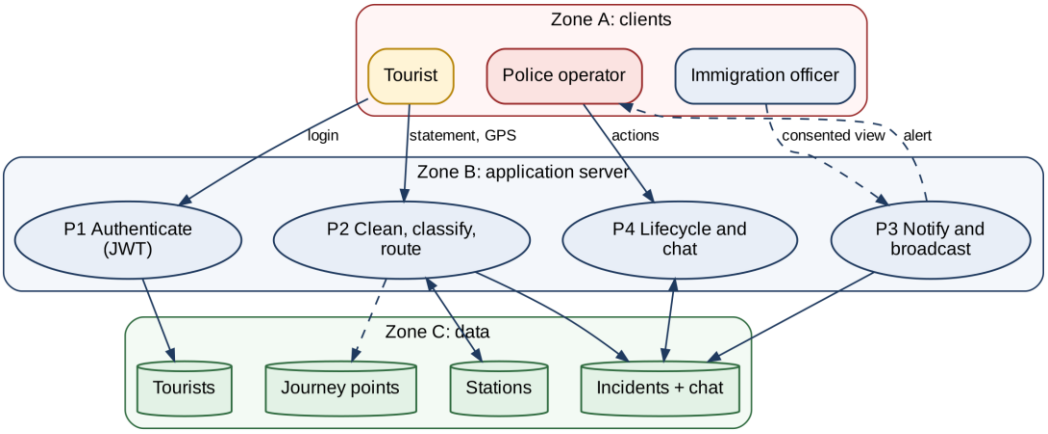


Figure 2.16: Data flow diagram showing the movement of personal data

Table 2.16: Threat analysis and controls

Threat	Impact	Control
Unauthorised access to incidents	Exposure of tourist data	JWT, role-based access, owner and station scoping (implemented); MFA (planned)
False or repeated SOS alerts	Wasted police resources	Authenticated users only, duplicate detection, warning against false reports
Misclassification or mistranslation	Wrong or delayed response	Original statement always shown, confidence shown, operator can override, fail-safe delivery
Location inaccuracy or denied GPS	Responders sent to wrong place	Accuracy radius shown; SOS blocked with guidance and hotlines if GPS is denied
Loss of connectivity	Alert not sent	Offline queue with duplicate-safe replay
Excessive data retention	Privacy violation	Retention schedule (planned)

2.8.2 Privacy and legal considerations

The system processes personal data of foreign visitors. The Personal Data Protection Act No. 9 of 2022 [44] establishes obligations for processing personal data, and the Data Protection Authority is the regulator [45]. The design follows principles that these frameworks share: location and profile are shared only with the tourist's consent at the moment of the SOS trigger, the immigration view shows only tourist-consented events and there is no covert tracking, only the minimum personal data needed for response is stored, and the limits of AI classification are disclosed to users. A formal data-protection impact assessment is future work, and cross-border processing by cloud speech or translation services must be examined before any live deployment. This thesis does not give legal advice.

2.8.3 Ethical considerations

- **Training data.** The corpus is synthetic or paraphrased and contains no real victims' words or personal data.
- **Advisory AI.** Classification is a suggestion. Operators see the confidence score and the original statement and can override the label. This counters automation bias.
- **Consent and surveillance.** Location is collected only when the tourist triggers an SOS or enables Safe Journey.

- **Fairness across languages.** An English-centric model may serve English speakers better than others. Translation reduces but does not remove the risk.
- **Official channels.** The interface keeps 119 and 1912 visible, warns against false reports and does not impersonate the official services.
- **Honest reporting.** Results that are less favourable to the deployed model (Section 3.1.3) are reported in full.
- **Human participants.** No participants were recruited for data collection. Any future usability study requires informed consent and ethical clearance [TBD: state whether ethical clearance was needed or obtained for the stakeholder reviews].

2.8.4 Social impact

Positive impacts are faster help for tourists, reduced language barriers and a digital evidence trail that supports prompt reporting. Possible negative impacts are false confidence in an unproven system, unequal service for poorly supported languages and misuse of location data. The work supports Sustainable Development Goals 3, 8, 9 and 11 [59] but does not claim to have achieved them.

2.8.5 Limitations

- The corpus is small (264 statements, 14 classes) and synthetic, so metrics are unstable and may not carry over to real speech.
- The output is single-label, so overlapping incidents are forced into one type.
- The vectoriser is English-centric and depends on translation quality for other languages.
- Straight-line distance is not travel time; stations in development are simulated seed data.
- Browser GPS can be inaccurate indoors or if permission is denied.
- There is no live integration with Police operational systems, so real end-to-end delay is unknown.

2.9 Commercialisation

2.9.1 Value proposition and market

The value proposition is safer, clearer emergencies for inbound tourists and faster, structured alerts for the Tourism Police, with AI explained in plain language on the SOS screens. Sri Lanka recorded about 2.36 million tourist arrivals in 2025 [3], and the authorities have shown interest in tourist security by activating dedicated channels such as the 1997 hotline [5]. Competing options are generic map SOS pins, hotel concierge calls and direct dialling of 119; none combines TourCeylon identity, text classification and journey replay. The module is positioned as an integrated safety layer within a tourism platform and not as a standalone emergency app, which lowers the barrier to adoption: tourists who already use the platform for planning have SOS available without installing anything new. A revenue-based market size has not been estimated from primary data [TBD].

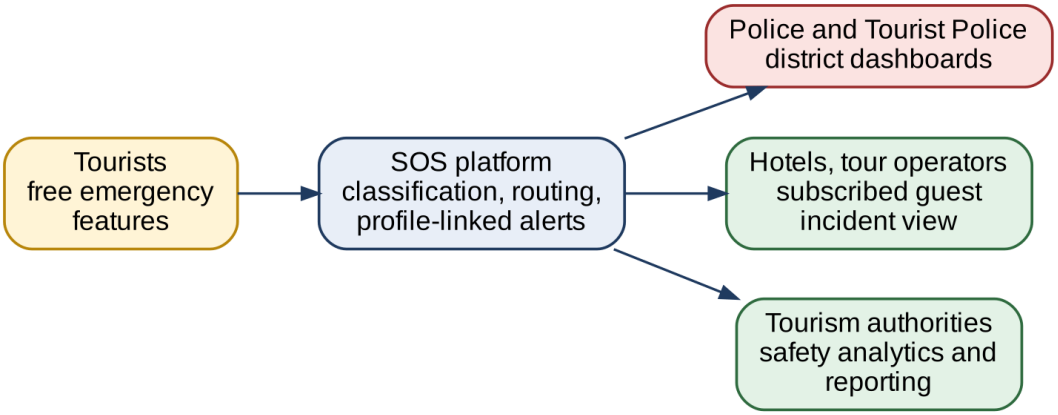


Figure 2.17: Value chain of the proposed SOS platform

Table 2.17: Stakeholders and benefits

Stakeholder	Benefit
Tourists	One-tap help with automatic location, in their own words or voice
Tourism Police	Structured alerts with type, urgency, map and profile; fewer clarifying questions
Medical and fire services	Correctly routed priority-one alerts with exact location
SLTDA	Digital complement to 1912; data on incident patterns for policy
Hotels and tour operators	Duty-of-care support for guests
TourCeylon platform	Trust feature that differentiates the platform

2.9.2 Business and sustainability model

Because an emergency feature should never be paywalled for tourists, revenue would come from institutional and business partners, while the cost of operation remains low because classification uses a small model on standard servers. Table 2.18 lists the candidate options. They are not validated by customer commitments.

Table 2.18: Candidate sustainability and revenue options (not validated)

Option	Description
Platform feature	SOS included free for all TourCeylon users as a trust-building feature that drives platform adoption
Institutional partnership	Hosted service for SLTDA and the Tourism Police with 1912 cross-referral messaging
Business duty-of-care subscription	Subscription for hotels and tour operators to monitor and support guests' emergencies
Academic demonstration	SLIIT-hosted demonstration back end for research and pilots

A numerical financial forecast is not given because no cost data or customer commitments exist [TBD]. Costs would include hosting, SMS and translation usage charges (to be checked against the current provider price lists), maintenance and support.

2.9.3 SWOT analysis

Table 2.19: SWOT analysis of the SOS platform

Strengths	Weaknesses
Integrated with a tourism platform; GPS-backed structured briefings; fail-safe delivery; offline queue; transparent evaluation	Small synthetic corpus; Random Forest weak on fourteen classes; no live police integration; English-centric model
Opportunities	Threats
Growing arrivals; government interest in tourist safety; partnerships with hotels and insurers; larger corpus from pilots	Privacy and legal constraints; institutional adoption delays; competing global SOS features in phones

2.9.4 Scalability, intellectual property, compliance and pilot plan

Flask-SocketIO can be scaled with multiple workers and a message queue, and the stateless classification step scales horizontally. The corpus, training pipeline, briefing design and integration code constitute the main intellectual property of the group. Compliance requirements include provisions against false reports, consent-based

processing of location and profile data under the Personal Data Protection Act [44] and GDPR-style principles for international visitors, and clear statements that the module complements and does not impersonate 119 and 1912.

A pilot would proceed in stages (Figure 2.18): consultation with Police representatives, pilot design in one or two areas, technical integration with station channels, field testing with simulated alerts, and phased expansion. Each stage would include a data-protection impact assessment and agreements on data retention. The author reports having contacted relevant stakeholders and obtained letters associated with the Police Headquarters and immigration authorities, and having submitted the research proposal [TBD: state issuer, date and exact scope of each letter]. These must not be read as approval for live deployment.

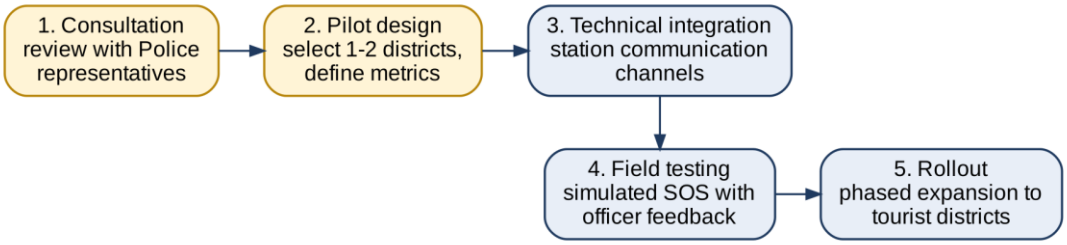


Figure 2.18: Proposed phased pilot and roll-out

2.9.5 Contribution to the Sustainable Development Goals

The module contributes to SDG 3 (good health and well-being) through faster medical emergency response, SDG 8 (decent work and economic growth) by supporting a safe tourism economy, SDG 9 (industry, innovation and infrastructure) through digital public-safety infrastructure, and SDG 11 (sustainable cities and communities) by making tourist routes safer [59].

2.10 Testing and Implementation

2.10.1 Test strategy

Verification combined unit tests for classification and statement cleaning, an authorisation matrix, duplicate-incident detection, multilingual communication tests, a tourist alerts-feed test and manual GPS-denial scenarios (Figure 2.19). The automated tests are in tests/test_sos_ml.py,

test_priority_one_dispatch_and_duplicates.py, test_authorization_matrix.py and test_tourist_alerts_feed.py.

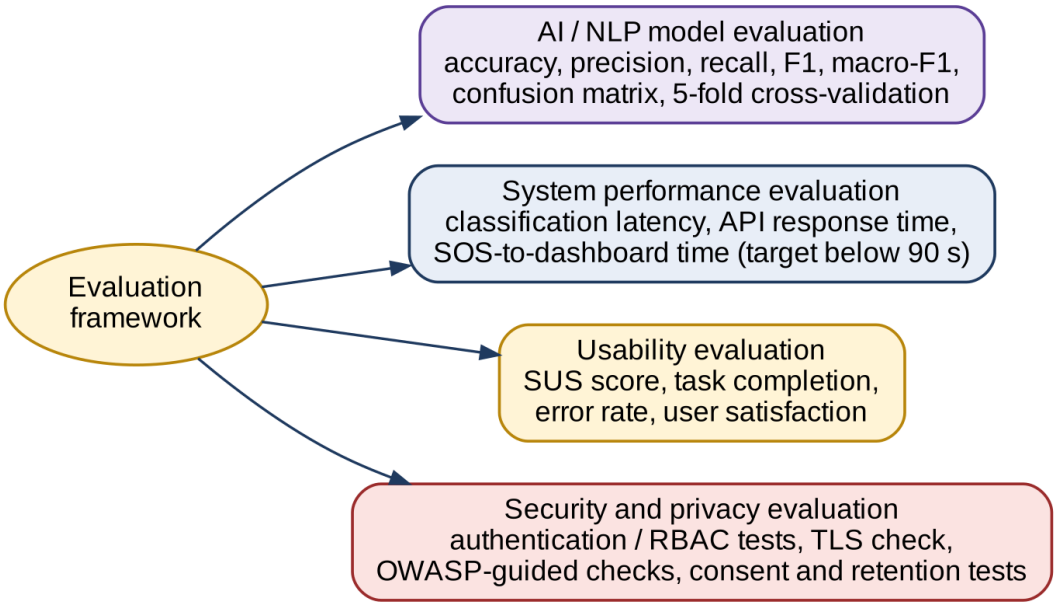


Figure 2.19: Evaluation framework

2.10.2 Functional test cases

Table 2.20: Representative test cases

ID	Test	Expected result	Result
T1	Classify a clear theft statement (test_sos_ml.py)	Theft predicted; responder police	[Pass/Fail]
T2	Classify with model bundle unavailable	Keyword fallback label returned; incident still created	[Pass/Fail]
T3	Statement cleaning	Filler words and duplicates removed; meaning preserved	[Pass/Fail]
T4	Priority-one dispatch	Medical or fire incident triggers immediate dispatch notification	[Pass/Fail]
T5	Duplicate SOS within a short interval	Second alert ignored or merged	[Pass/Fail]
T6	Authorisation matrix	Each role accesses only permitted endpoints	[Pass/Fail]
T7	SOS without GPS	Request rejected with guidance; hotlines displayed	[Pass/Fail]
T8	Offline SOS and reconnect	Queued alert replayed once; incident visible on dashboard	[Pass/Fail]
T9	Multilingual chat message	Message translated and delivered	[Pass/Fail]
T10	TourCeylon login handoff	SOS opened without second login	[Pass/Fail]
T11	Tourist alerts feed (test_tourist_alerts_feed.py)	Police feed shows cleaned tourist statement, not template noise	[Pass/Fail]

2.10.3 Scenario walkthroughs

Six end-to-end scenarios are walked through and recorded (Table 2.21). Each checks that the GPS position is captured, the alert is delivered, the briefing is complete and, where relevant, that offline replay works.

Table 2.21: Scenario walkthrough plan

Scenario	Checks
Phone snatched at a railway station (text)	GPS, police routing, briefing, chat
Companion collapses on a hiking trail (voice)	Speech-to-text, medical routing, priority-one dispatch
Harassment on a beach (text, translated)	Translation, police routing, map link
Guesthouse kitchen fire (text)	Fire routing, priority-one dispatch
Tuk-tuk crash in an area with no signal	Offline queue, replay, duplicate safety
GPS permission denied	SOS blocked with guidance; 119 and 1912 shown

2.10.4 Performance, usability and field evaluation

Latency was not measured. Timings would be recorded at four points (SOS trigger, transcript available, classification complete, alert visible on the dashboard) over at least 30 repetitions, reporting the median, 95th percentile and maximum. Usability so far was assessed by supervisor and stakeholder review of the screens and briefing format. A System Usability Scale study [60], interpreted with the adjective ratings of Bangor et al. [61], is planned; small groups reveal most usability problems [43] but a larger group is needed for a precise SUS estimate. A comparison of the information available to a responder under a voice-only call and under the SOS alert is structural and is presented in Section 3.1.5; a timed field comparison would require ethical approval and partner participation.

2.10.5 Screenshots of the implemented system

The following figures are placeholders for screenshots of the working system. They must be replaced with real screenshots.

[INSERT FIGURE 2.20 HERE]

Screenshot of SOSPage

Figure 2.20: Tourist SOS screen with GPS status, statement input and hotline numbers

[INSERT FIGURE 2.21 HERE]

Screenshot of the offline state

Figure 2.21: Queued alert message shown when the network is unavailable

[INSERT FIGURE 2.22 HERE]

Screenshot of the dashboard list and map

Figure 2.22: Police dashboard with incoming incidents and map

[INSERT FIGURE 2.23 HERE]

Screenshot of the briefing message

Figure 2.23: Structured briefing in the incident chat

[INSERT FIGURE 2.24 HERE]

Screenshot of the immigration view

Figure 2.24: Immigration read-only view of consented events

2.10.6 Integration with the other components

The TourCeylon itinerary client opens the SOS module through an access-token exchange (routes/tourists.py, sosClient.js), so a tourist who is logged into the planning application does not need separate SOS credentials. The shared PostgreSQL database stores tourists and incidents. The risk-intelligence and driver modules may consume incident context in future enhancements, for example to warn other travellers about an area or to alert a driver assigned to the tourist.

2.11 Tools and Environment

Table 2.22: Tools used in the research

Purpose	Tool
Programming	Python 3, JavaScript (React)
Machine learning	scikit-learn, joblib
Back end and real time	Flask, Flask-SocketIO, SQLAlchemy

Purpose	Tool
Database	PostgreSQL
Retraining	train_sos_ml.py, RETRAIN_SOS_ML.bat
Diagrams	Graphviz and Matplotlib
Writing	Microsoft Word; generative AI assistance disclosed in Appendix F

3. RESULTS AND DISCUSSION

This chapter reports what was measured. The measured results are the hold-out metrics recorded by the training script in metrics.json (264 statements, 14 incident classes, stratified 80/20 split, about 53 test statements). Everything else is reported as implemented, tested by automated and scenario tests, or not yet evaluated. Items marked in the text are values that the author must copy from metrics.json or from the final test run. Table 3.1 summarises which claims the evidence supports.

Table 3.1: Evidence status of the main claims

Claim	Status	Evidence
The tourist client, police dashboard and immigration view are implemented	Supported	Feature list (Table 3.2), screenshots
Random Forest responder routing is usable	Supported on hold-out (accuracy ≈ 0.811)	metrics.json
Random Forest beats keyword rules on 14 incident types	Not supported (macro-F1 ≈ 0.359 vs ≈ 0.496)	metrics.json
Logistic Regression beats keyword rules on 14 incident types	Supported on hold-out (macro-F1 ≈ 0.707)	metrics.json
Statistical significance of model differences	Not tested	Planned (Section 2.4.6)
The system supports ten languages	Design target only	No per-language evaluation
Detection-to-dispatch under 90 s	Proposal target; not measured	None
Deployed with, or integrated into, police systems	Not claimed	None

3.1 Results

3.1.1 Implementation status and functional testing

Table 3.2 lists the delivered features. The automated tests and scenario walkthroughs of Section 2.10 are recorded in Table 3.3.

Table 3.2: Product features delivered

Feature	Status
One-tap SOS with mandatory GPS	Implemented
Text and voice input (speech-to-text, translation, audio preservation)	Implemented
AI incident type, confidence and urgency; keyword fallback	Implemented
Responder routing to the nearest station	Implemented
SocketIO dashboard updates and SMS or e-mail notifications	Implemented

Feature	Status
Structured police briefing with map links	Implemented
Safe Journey route history	Implemented
Offline queue with duplicate-safe replay	Implemented
Police acknowledge, escalate and close; immigration read-only view	Implemented
TourCeylon login handoff	Implemented

Table 3.3: Results of the representative test cases (to be completed from the final test run)

Test	Result	Test	Result
T1 Clear theft statement	[Pass/Fail]	T7 SOS without GPS	[Pass/Fail]
T2 Bundle unavailable (fallback)	[Pass/Fail]	T8 Offline SOS and replay	[Pass/Fail]
T3 Statement cleaning	[Pass/Fail]	T9 Multilingual chat	[Pass/Fail]
T4 Priority-one dispatch	[Pass/Fail]	T10 Login handoff	[Pass/Fail]
T5 Duplicate SOS	[Pass/Fail]	T11 Alerts feed	[Pass/Fail]
T6 Authorisation matrix	[Pass/Fail]		

The offline queue replayed successfully in the test scenarios, and the voice pipeline preserved the audio when speech-to-text failed [confirm from test run].

3.1.2 Dataset description

The dataset record is in Table 2.3. The corpus has 264 labelled synthetic or paraphrased statements across 14 incident types and 3 responder categories. Because 264 statements are shared across 14 classes, several classes have only a few examples, which strongly influences the results below. The count per class is given in Appendix A [insert counts].

3.1.3 Model comparison

Table 3.4 reports the hold-out results for the incident-type task and Table 3.5 for the responder task. Values are rounded and are taken from metrics.json. The figure Figure 3.1 and Figure 3.2 show the macro-F1 values.

Table 3.4: Incident-type classification, 14 classes, hold-out set

Model	Accuracy	Macro-F1
Dummy (most frequent class)	not recorded	≈ 0.012
Keyword rules (TYPE_KEYWORDS)	≈ 0.547	≈ 0.496

Model	Accuracy	Macro-F1
Random Forest (production)	[insert from metrics.json]	≈ 0.359
Logistic Regression (candidate)	[insert from metrics.json]	≈ 0.707

Table 3.5: Responder classification, 3 classes, hold-out set

Model	Accuracy	Macro-F1
Dummy (most frequent class)	not recorded	≈ 0.265
Random Forest	≈ 0.811	≈ 0.684

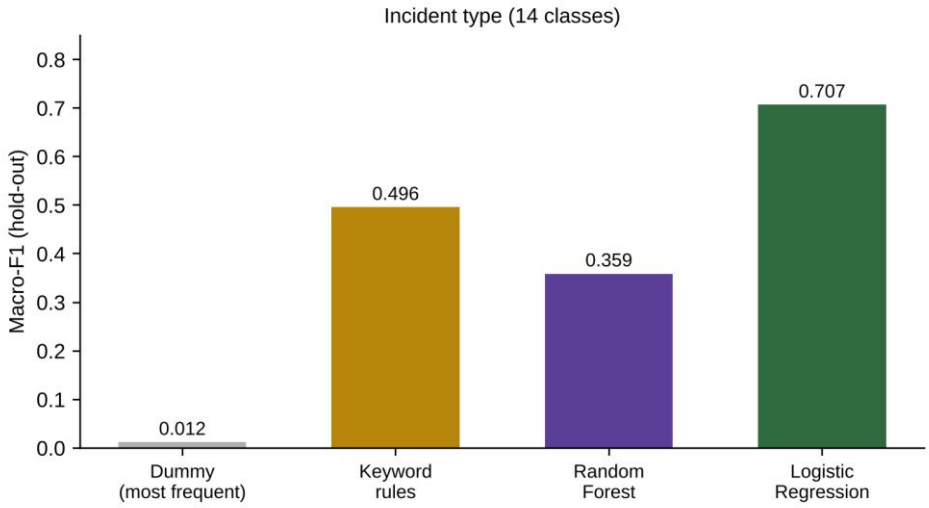


Figure 3.1: Hold-out macro-F1 for incident-type classification (14 classes)

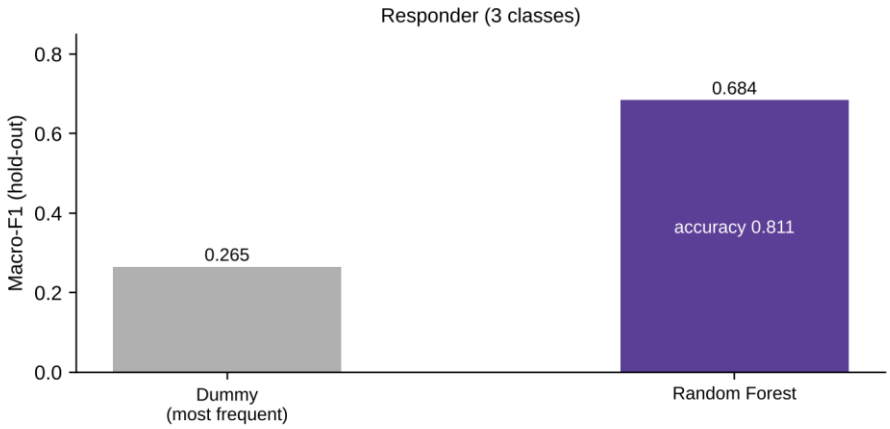


Figure 3.2: Hold-out macro-F1 for responder classification (3 classes)

3.1.3.1 Incident-type classification

On fine-grained fourteen-way classification, the Random Forest reached a hold-out macro-F1 of about 0.359. This is far above the dummy baseline (about 0.012) but below the keyword baseline (about 0.496) and below the Logistic Regression candidate (about 0.707). The principal cause is sparse per-class support. With 264 statements across 14 classes, many classes contribute only a handful of examples to the hold-out set, so a few misclassifications sharply lower their F1, and the Random Forest has too few examples per class to learn reliable splits on sparse TF-IDF features. Logistic Regression, which spreads small weights across many terms, is better suited to this regime, in line with the literature on short texts [25].

The keyword baseline performs relatively well on this corpus because many statements contain a strongly indicative word, such as stolen, fire or bleeding, that the keyword list captures directly. Keyword rules fail mainly on paraphrases that avoid these words, while the Random Forest fails mainly on rare classes. The two error patterns are partly complementary, which suggests that a hybrid that uses keywords for high-precision cues and a learned model for the rest could outperform either alone. The production system nevertheless serves the Random Forest, refitted on all labels, for consistency with the Random Forest components of the group platform, with keyword fallback if the bundle is not loaded; the implications are discussed in Section 3.2.1. The thesis does not claim that the Random Forest beats keyword rules across all 14 classes.

3.1.3.2 Responder routing

The responder Random Forest achieved hold-out accuracy of about 0.811 and macro-F1 of about 0.684, much higher than the dummy responder baseline (macro-F1 about 0.265). With three classes and many more examples per class, the Random Forest can learn the vocabulary that distinguishes medical situations (injury, breathing, bleeding, collapsed) from fire (smoke, burning) and police matters (stolen, following, attacked). Routing is therefore usable even when fine-grained type labels are sparse. The responder model decides the most important routing question in an emergency, namely which kind of service is alerted first.

3.1.3.3 Error analysis

Inspection of misclassifications shows three recurring patterns. First, overlapping incidents: a statement such as “they attacked us and took my phone, my friend is hurt” legitimately involves assault, theft and injury, and any single label is partly wrong. Second, rare classes with few training statements are often predicted as a semantically neighbouring common class. Third, very short statements such as “help please quickly” contain no incident-specific words, so every model, including keyword rules, can only guess. The first pattern suggests that multi-label output may suit the domain better; the second and third indicate that more and more varied data per class would help most. Per-class precision, recall and F1 are in Appendix A [insert per-class table from metrics.json].

3.1.4 System performance

Response time was not measured. The proposal target of under 90 seconds from detection to dispatch is therefore reported as not yet evaluated. Classification itself is a single TF-IDF transform and two Random Forest predictions on a small model bundle, and the non-functional target is sub-second classification; a timing measurement following the plan in Section 2.10.4 is needed before this can be reported as a result [insert measured classification time if available].

3.1.5 Responder information, scenarios and stakeholder validation

Information available to the responder. RQ3 asks whether a GPS-anchored alert with profile and briefing gives responders more information at the start of the response than a voice-only call. Table 3.6 compares the two. The comparison is structural and not a timed field study, but it shows that the digital alert removes the questions that typically take the first part of an emergency call.

Table 3.6: Information available at the start of the response

Information	Voice-only 119 call	TourCeylon SOS
Exact location	Verbal description	GPS coordinates with map link and accuracy
Movement or route	Verbal	Live updates and Safe Journey history
Identity and nationality	Must be asked	From the shared profile
Incident type	Operator interprets	AI label with confidence and the original statement

Information	Voice-only 119 call	TourCeylon SOS
Urgency	Operator judgement	Mapped from type; priority-one flagged
Language barrier	Depends on operator	Translation of text and chat
Written record	Call log only	Structured incident record and chat history

Scenario walkthrough. A representative scenario shows the product result. A tourist on a train in the hill country types “a man snatched my phone and pushed me, I am at the station now”. The client captures GPS and sends the statement. The model labels the incident as theft-related with a confidence score, the urgency mapping assigns a level, and the responder model selects police. Routing selects the nearest seeded police station, whose dashboard receives the incident in real time, and the chat opens with a briefing containing the type, confidence, cleaned statement, name and nationality, and a map link. The officer acknowledges the incident, the tourist sees the status change, and the two continue in chat. In contrast to a voice call, the officer does not need to ask where the tourist is or who they are. Table 3.7 records the outcome of the six scenarios.

Table 3.7: Scenario walkthrough outcomes (to be completed)

Scenario	Outcome
Phone snatched at a railway station (text)	[record outcome]
Companion collapses on a hiking trail (voice)	[record outcome]
Harassment on a beach (text, translated)	[record outcome]
Guesthouse kitchen fire (text)	[record outcome]
Tuk-tuk crash in an area with no signal	[record outcome]
GPS permission denied	[record outcome]

Stakeholder validation. Supervisor and stakeholder review focused on the clarity of the AI explanation shown to users, the mandatory GPS policy and alignment with the Tourist Police’s encouragement of prompt reporting [13]. Reviewers valued that the original statement is always shown next to the AI label, so that an operator is never forced to rely on the model alone, and recommended keeping 119 and 1912 visible on every SOS screen, which was implemented [insert any additional reviewer or user feedback with dates]. A System Usability Scale study has not yet been carried out, so no SUS score is reported.

[INSERT FIGURE 3.3 HERE]

Insert the pytest output

Figure 3.3: Output of the automated test run

3.1.6 Security and privacy checks

The security measures of Section 2.8 are verified by the authorisation-matrix test (T6), the duplicate-incident test (T5) and the GPS-requirement test (T7); the results are those of Table 3.3. Penetration or load testing was not carried out.

3.1.7 Integration and institutional engagement

The JSON Web Token handoff from the TourCeylon login succeeded in bootstrap tests, and itinerary-module links do not require separate SOS credentials. Incidents reference the shared tourist identity, so profile information entered once in TourCeylon is available in the police briefing [confirm]. The author reports having contacted relevant stakeholders, having obtained permission or approval letters associated with the Police Headquarters and immigration authorities, and having submitted the proposal. Table 3.8 separates the possible types of approval.

Table 3.8: Institutional engagement record (to be completed)

Type	Status
Proposal submitted	Reported; date [TBD]
Letter received	Reported; issuer, date and wording [TBD]
Permission for a type of testing	[TBD]
Access to restricted data or systems	Not claimed
Approval for live operational deployment	Not claimed

3.2 Discussion

3.2.1 Interpretation of findings

Machine-learning gains are largest on responder routing and on paraphrases compared with static keywords; sparse classes need more labelled examples per type. The responder model's accuracy of about 0.81 means that in most cases the alert reaches the correct type of station automatically.

The fourteen-class result requires careful interpretation. The Random Forest's macro-F1 of about 0.359 is lower than the keyword baseline's 0.496, so RQ1 is not supported for the Random Forest on the current corpus. However, the Logistic Regression candidate, trained on exactly the same TF-IDF features, reached about 0.707, clearly above the keywords. Learned classification is therefore supported in general but not for the specific algorithm currently served. The bottleneck is the interaction between algorithm and data volume and not the idea of learned classification. The decision to serve the Random Forest was taken for consistency with the group's Random Forest itinerary and risk modules and for its robustness in the shared stack. In a safety-critical setting, however, predictive quality should dominate. Because the bundle format is independent of the classifier, switching the incident-type model to Logistic Regression is a small change in the training script, and this is recommended as the first item of future work. Neither choice affects alert delivery: GPS, routing to a station and notification occur regardless of the label. These results are indicative because the hold-out set is small and no significance test has been carried out.

3.2.2 Answers to the research questions

Table 3.9: Answers to the research questions

Question	Answer	Evidence
RQ1 Random Forest versus keywords and Logistic Regression (incident type)	Not supported for Random Forest; supported for Logistic Regression	Macro-F1 ≈ 0.359 (RF), ≈ 0.496 (keywords), ≈ 0.707 (LR)
RQ2 Responder routing	Supported	Accuracy ≈ 0.811; macro-F1 ≈ 0.684 against dummy ≈ 0.265
RQ3 Information for responders	Supported (structural evidence)	Table 3.6; scenarios
RQ4 Reliable workflow	Supported by implementation and tests	Fallback, offline queue, duplicate control; test results to be inserted

Question	Answer	Evidence
RQ5 Acceptability	Partly addressed by stakeholder review; SUS not done	Section 3.1.5

3.2.3 Comparison with existing solutions and literature

Compared with calling 119 alone, the artefact adds incident type, urgency, maps and profile to every alert. Compared with generic chatbots, it uses supervised models trained on SOS vocabulary and does not generate details about the emergency. Compared with phone-level SOS features that share a location with contacts [15], it routes to responders and structures the information for them. Relative to crisis-informatics research on classifying emergency messages [17], this work applies similar ideas to a smaller, tourist-specific taxonomy and integrates classification into an operational workflow. A numerical comparison with published classifiers is not appropriate, because they were evaluated on public disaster posts and not on short first-person statements [18], [19].

3.2.4 Limitations and threats to validity

Table 3.10: Limitations and mitigation

Limitation	Effect	Mitigation
Small corpus (264 statements, 14 classes)	Unstable per-class metrics; Random Forest underfits rare classes	Expand the corpus; collect consented, de-identified real statements
Synthetic or paraphrased text	May not reflect real stressed language, slang or code-mixing	Validate with field pilots and Tourism Police input
English-centric vectoriser	Depends on translation quality for Sinhala, Tamil and other input	Multilingual features or models
Single-label output	Overlapping incidents forced into one type	Multi-label classification
Simulated stations in development	Routing not tested against real coverage	Seed real station data with partners
Browser GPS dependence	Indoor or denied permission reduces accuracy	Show accuracy radius; allow landmark text
Ethics limits on real incidents	Evaluation cannot use real emergencies	Scenario testing and expert review

Internal validity is threatened mainly by the small hold-out set, where a single statement can shift per-class F1 noticeably, and by the lack of cross-validation. External validity is limited because the corpus is synthetic and written by one author.

Construct validity is supported by standard metrics and by baselines, and conclusion validity by a fixed stratified split and a stored metrics file; no significance test was run.

3.2.5 Ethical reflection

Building an AI component for emergencies raises ethical questions beyond accuracy. The first is consent and surveillance: location tracking can protect but can also be misused, so the module collects location only when the tourist triggers an SOS or enables Safe Journey, and the immigration view displays only tourist-consented events. The second is automation bias: operators may trust an AI label too much, so the confidence and the original statement are always shown and the label is worded as a suggestion. The third is fairness across languages. Finally, publishing results that are less favourable to the deployed model is itself an ethical choice, because it allows examiners and future developers to make informed decisions.

3.2.6 Lessons learned and implications

Three practical lessons emerged. First, delivery must never depend on AI: separating alert creation from classification ensured that model failures could only degrade the briefing, not lose an alert. Second, baselines are essential: a single headline metric would have hidden the fact that a simpler model is better on this data. Third, presentation matters as much as prediction: cleaning statements and consolidating the briefing made the output usable to officers, whereas raw labels and transcripts were not. For authorities, a structured digital alert reduces the time spent establishing basic facts, and public voice hotlines remain complementary channels.

3.3 Summary of Student Contribution

TourCeylon is a group project in which each member is responsible for one component. This thesis reports the individual work of De Silva D.S.K (IT22108654) on the SOS Emergency Safety System. The shared Flask and PostgreSQL infrastructure is group-owned. The machine-learning design and the SOS-specific routes are individual deliverables. Table 3.11 summarises the contribution. The other components of the group (itinerary planning, wellness matching, crowd prediction and risk intelligence) are described in their own theses and are not claimed here.

Table 3.11: Summary of the individual contribution

Area	Contribution of the author
Research	Literature review, gap analysis, research questions and objectives for tourist emergency response
Data and AI	Corpus of 264 labelled statements; training pipeline; comparison of Random Forest, Logistic Regression, keyword and dummy baselines; sos_models.joblib and metrics.json
Back end	SOS data models; incident create, classify and routing services; urgency mapping; notifications; briefing; offline replay handling; TourCeylon login handoff
Front end	Tourist SOS client with GPS, voice and text, offline queue and Safe Journey; police dashboard; immigration view
Evaluation	Hold-out evaluation, automated tests, scenario walkthroughs, stakeholder review
Stakeholder engagement	Contact with Police Headquarters and immigration-related officials; proposal submission (as reported)

4. CONCLUSION

4.1 Summary of the Research

This thesis addressed the problem that a foreign tourist in Sri Lanka may find it hard to reach the right emergency channel, to say where they are and to describe what has happened, often in a second language, under stress and with unreliable connectivity. It reviewed the tourism safety, emergency-communication, text-classification, speech and data-protection literature together with public reports, identified five research gaps, and presented the SOS Emergency Safety System of the TourCeylon platform. The system lets an authenticated tourist raise a GPS-anchored SOS by text or voice, classifies the statement with TF-IDF and Random Forest models, routes the alert to the nearest suitable station, pushes it to police and immigration dashboards in real time with a structured briefing, and queues alerts offline.

The evaluation was a hold-out comparison on a corpus of 264 synthetic statements. The responder Random Forest reached an accuracy of about 0.811 and a macro-F1 of about 0.684, far above a dummy baseline. On the fourteen-class incident task the Random Forest reached a macro-F1 of about 0.359, below a keyword baseline (about 0.496) and a Logistic Regression candidate (about 0.707). The thesis reports this transparently. It concludes that structured, map-backed digital SOS with machine-learning triage complements, and does not replace, national voice hotlines.

4.2 Achievement of the Objectives

Table 4.1: Status of the specific objectives

Objective	Status	Evidence
1 Requirements	Achieved	Sections 2.1, 2.6
2 Dataset	Achieved with limits: 264 synthetic statements, 14 types	Sections 2.3, 3.1.2
3 Model comparison	Achieved: hold-out comparison; significance test and cross-validation planned	Sections 2.4, 3.1.3
4 Tourist application	Achieved	Sections 2.5, 3.1.1
5 Nearest-station routing and offline queue	Achieved; test results to be inserted	Sections 2.5.5, 2.5.8
6 Police and immigration dashboards	Achieved	Section 2.5
7 Voice and translation path	Implemented; per-language accuracy not evaluated	Section 2.4.7

Objective	Status	Evidence
8 Security and privacy	Achieved at design and test level; MFA, audit log and retention planned	Section 2.8
9 Evaluation	Partly achieved: hold-out metrics and scenarios; no timing or SUS study	Chapter 3

Objectives 1 to 6 were met. Objectives 3, 7 and 9 were met in part, with the limitations noted above, notably the weakness of the Random Forest on the fourteen-class task.

4.3 Future Work

Table 4.2: Future work roadmap

Priority	Work item	Expected benefit
1	Serve Logistic Regression, or an ensemble, for incident type	Immediate gain in fourteen-class macro-F1 (≈ 0.707 against ≈ 0.359 on the hold-out set)
2	Expand the corpus per class; add Sinhala, Tamil and code-mixed statements; run cross-validation and McNemar's test	More stable metrics; fairer service across languages
3	Multi-label incident output	Correct representation of overlapping incidents
4	Pilot with the Tourism Police using real station data; timed response and SUS study	External validity; real routing coverage
5	Integration APIs with 1912 and 119; multi-factor authentication, audit log, retention schedule and data-protection impact assessment	Seamless hand-over and stronger governance
6	On-device speech recognition; progressive or native mobile client	Voice SOS that works without connectivity

4.4 Concluding Remarks

The hardest parts of a tourist emergency, namely saying where you are, who you are and what kind of help you need, can be largely automated with established web technologies and lightweight machine learning, while the final judgement stays with trained responders. The evaluation also shows the value of measuring carefully: it identified exactly where the current models fall short and how to improve them. The module is therefore both a working prototype and a documented foundation for an operational tourist safety service in Sri Lanka.

REFERENCES

[1] A. Pizam and Y. Mansfeld, Eds., *Tourism, Security and Safety: From Theory to Practice*. Oxford, U.K.: Butterworth-Heinemann, 2006.

[2] S. F. Sönmez and A. R. Graefe, "Influence of terrorism risk on foreign tourism decisions," *Ann. Tourism Res.*, vol. 25, no. 1, pp. 112–144, 1998.

[3] "Sri Lanka logs all-time-high 2.36 million annual tourist arrivals in 2025; average tourist spending decreases," *Bangladesh Monitor*, Jan. 15, 2026. [Online]. Available: https://www.bangladeshmonitor.com.bd/lead-news-details/sri-lanka-logs-all-time-high-236-million-annual-tourist-arrivals-in-2025-average-tourist-spending-decreases (accessed Oct. 10, 2026).

[4] Sri Lanka Tourism Development Authority, "Dial 1912: Tourism hotline." [Online]. Available: https://sltda.gov.lk/en/dial-1912 (accessed Oct. 10, 2026).

[5] Z. Farzan, "Sri Lanka boosts tourist security amid global tensions; hotline 1997 now active," *News 1st*, Oct. 23, 2024. [Online]. Available: https://english.newsfirst.lk/2024/10/23/sri-lanka-boosts-tourist-security-amid-global-tensions-hotline-1997-now-active (accessed Oct. 10, 2026).

[6] "A new mobile app for safety of tourists – Harin," *News 1st*, Dec. 5, 2022. [Online]. Available: https://www.newsfirst.lk/2022/12/05/a-new-mobile-app-for-safety-of-tourists-harin/ (accessed Oct. 10, 2026).

[7] TripAdvisor, "Trip report March/April 2025," Sri Lanka Forum. [Online]. Available: https://www.tripadvisor.co.nz/ShowTopic-g293961-i8983-k15315033-Trip_report_March_April_2025-Sri_Lanka.html (accessed Oct. 2026).

[8] Sunday Times (Sri Lanka), "Tourism: Perils of playing it by ear!," *Business Times*, Mar. 30, 2025. [Online]. Available: https://www.sundaytimes.lk/250330/business-times/tourism-perils-of-playing-it-by-ear-2-593418.html (accessed Oct. 2026).

[9] Israel National News, "Jerusalem traveler survives violent robbery in Sri Lanka," Aug. 18, 2025. [Online]. Available: https://www.israelnationalnews.com/news/413454 (accessed Oct. 2026).

[10] The Morning Telegraph, "Spanish tourist attacked on Sri Lankan train, phone recovered by police," Dec. 2025. [Online]. Available: https://themorningtelegraph.com/17820/ (accessed Oct. 2026).

[11] Daily Mirror, "Man arrested after video of tourist harassment goes viral," Nov. 16, 2025. [Online]. Available: https://www.dailymirror.lk/breaking-news/Man-arrested-after-video-of-tourist-harassment-goes-viral/108-325145 (accessed Oct. 2026).

[12] Ceylon Today, "Harassment of Dutch tourists: Suspect handed 10-year suspended sentence," Nov. 25, 2025. [Online]. Available: https://ceylontoday.lk/2025/11/25/harassment-of-dutch-tourists-suspect-handed-10-year-suspended-sentence/ (accessed Oct. 2026).

[13] The Morning, "Ensuring tourist safety amidst growth," 2025. [Online]. Available: https://www.themorning.lk/articles/2DsN2x9djuRf5HO5MSRU (accessed Oct. 2026).

[14] Sri Lanka Police, "In an emergency that requires immediate response, call the emergency number 119," official announcement. [Online]. Available: https://www.police.lk/?p=18161 (accessed Oct. 2026).

[15] Google, "Emergency SOS on Pixel phones," Pixel Phone Help. [Online]. Available: https://support.google.com/pixelphone/answer/7055029 (accessed Oct. 10, 2026).

[16] C. Castillo, *Big Crisis Data: Social Media in Disasters and Time-Critical Situations*. New York, NY, USA: Cambridge Univ. Press, 2016.

[17] M. Imran, C. Castillo, F. Diaz, and S. Vieweg, "Processing social media messages in mass emergency: A survey," *ACM Comput. Surv.*, vol. 47, no. 4, Art. no. 67, 2015.

[18] M. Imran, P. Mitra, and C. Castillo, "Twitter as a lifeline: Human-annotated Twitter corpora for NLP of crisis-related messages," in *Proc. 10th Int. Conf. Lang. Resour. Eval. (LREC)*, 2016, pp. 1638–1643.

[19] A. Olteanu, C. Castillo, F. Diaz, and S. Vieweg, "CrisisLex: A lexicon for collecting and filtering microblogged communications in crises," in *Proc. 8th Int. AAAI Conf. Weblogs Social Media (ICWSM)*, 2014, pp. 376–385.

[20] T. Sakaki, M. Okazaki, and Y. Matsuo, "Earthquake shakes Twitter users: Real-time event detection by social sensors," in *Proc. 19th Int. Conf. World Wide Web (WWW)*, Raleigh, NC, USA, 2010, pp. 851–860.

[21] S. N. Blomberg *et al.*, "Effect of machine learning on dispatcher recognition of out-of-hospital cardiac arrest during calls to emergency medical services: A randomized clinical trial," *JAMA Netw. Open*, vol. 4, no. 1, Art. no. e2032320, 2021.

[22] F. Sebastiani, "Machine learning in automated text categorization," *ACM Comput. Surv.*, vol. 34, no. 1, pp. 1–47, 2002.

[23] G. Salton and C. Buckley, "Term-weighting approaches in automatic text retrieval," *Inf. Process. Manage.*, vol. 24, no. 5, pp. 513–523, 1988.

[24] T. Joachims, "Text categorization with support vector machines: Learning with many relevant features," in *Proc. 10th Eur. Conf. Mach. Learn. (ECML)*, 1998, pp. 137–142.

[25] S. Wang and C. D. Manning, "Baselines and bigrams: Simple, good sentiment and topic classification," in *Proc. 50th Annu. Meeting Assoc. Comput. Linguistics (ACL)*, 2012, pp. 90–94.

[26] L. Breiman, "Random forests," *Mach. Learn.*, vol. 45, no. 1, pp. 5–32, 2001.

[27] M. Fernández-Delgado, E. Cernadas, S. Barro, and D. Amorim, "Do we need hundreds of classifiers to solve real world classification problems?," *J. Mach. Learn. Res.*, vol. 15, pp. 3133–3181, 2014.

[28] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *J. Mach. Learn. Res.*, vol. 12, pp. 2825–2830, 2011.

[29] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.

[30] V. Sanh, L. Debut, J. Chaumond, and T. Wolf, "DistilBERT, a distilled version of BERT: Smaller, faster, cheaper and lighter," arXiv:1910.01108, 2019.

[31] T. Pires, E. Schlinger, and D. Garrette, "How multilingual is multilingual BERT?," in *Proc. 57th Annu. Meeting Assoc. Comput. Linguistics (ACL)*, Florence, Italy, 2019, pp. 4996–5001.

[32] A. Conneau *et al.*, "Unsupervised cross-lingual representation learning at scale," in *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics (ACL)*, 2020, pp. 8440–8451.

[33] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence embeddings using Siamese BERT-networks," in *Proc. EMNLP-IJCNLP*, 2019, pp. 3982–3992.

[34] N. Japkowicz and S. Stephen, "The class imbalance problem: A systematic study," *Intell. Data Anal.*, vol. 6, no. 5, pp. 429–449, 2002.

[35] M. Sokolova and G. Lapalme, "A systematic analysis of performance measures for classification tasks," *Inf. Process. Manage.*, vol. 45, no. 4, pp. 427–437, 2009.

[36] R. Kohavi, "A study of cross-validation and bootstrap for accuracy estimation and model selection," in *Proc. 14th Int. Joint Conf. Artif. Intell. (IJCAI)*, 1995, pp. 1137–1143.

[37] Q. McNemar, "Note on the sampling error of the difference between correlated proportions or percentages," *Psychometrika*, vol. 12, no. 2, pp. 153–157, 1947.

[38] T. G. Dietterich, "Approximate statistical tests for comparing supervised classification learning algorithms," *Neural Comput.*, vol. 10, no. 7, pp. 1895–1923, 1998.

[39] M. T. Ribeiro, S. Singh, and C. Guestrin, "\"Why should I trust you?\": Explaining the predictions of any classifier," in *Proc. 22nd ACM SIGKDD Int. Conf. Knowl. Discovery Data Mining*, 2016, pp. 1135–1144.

[40] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. 40th Int. Conf. Mach. Learn. (ICML)*, PMLR vol. 202, 2023, pp. 28492–28518.

[41] Y. Wu *et al.*, "Google's neural machine translation system: Bridging the gap between human and machine translation," arXiv:1609.08144, 2016.

[42] Google Cloud, "Cloud Translation documentation." [Online]. Available: https://cloud.google.com/translate/docs (accessed Oct. 10, 2026).

[43] R. W. Sinnott, "Virtues of the haversine," *Sky Telescope*, vol. 68, no. 2, p. 159, 1984.

[44] *Personal Data Protection Act, No. 9 of 2022*, Parliament of the Democratic Socialist Republic of Sri Lanka, 2022.

[45] Data Protection Authority of Sri Lanka, "Background." [Online]. Available: https://dpa.gov.lk/Background.php (accessed Oct. 10, 2026).

[46] P. A. Grassi *et al.*, "Digital identity guidelines: Authentication and lifecycle management," Nat. Inst. Standards Technol., Gaithersburg, MD, USA, NIST SP 800-63B, 2017.

[47] M. Jones, J. Bradley, and N. Sakimura, "JSON Web Token (JWT)," Internet Eng. Task Force, RFC 7519, May 2015.

[48] E. Rescorla, "The Transport Layer Security (TLS) protocol version 1.3," Internet Eng. Task Force, RFC 8446, Aug. 2018.

[49] OWASP Foundation, "OWASP Top 10:2021." [Online]. Available: https://owasp.org/Top10/ (accessed Oct. 10, 2026).

[50] *Information Security, Cybersecurity and Privacy Protection — Information Security Management Systems — Requirements*, ISO/IEC 27001:2022, 2022.

[51] A. R. Hevner, S. T. March, J. Park, and S. Ram, "Design science in information systems research," *MIS Quart.*, vol. 28, no. 1, pp. 75–105, 2004.

[52] K. Peffers, T. Tuunanen, M. A. Rothenberger, and S. Chatterjee, "A design science research methodology for information systems research," *J. Manage. Inf. Syst.*, vol. 24, no. 3, pp. 45–77, 2007.

[53] K. Beck *et al.*, "Manifesto for Agile Software Development," 2001. [Online]. Available: https://agilemanifesto.org/ (accessed Oct. 10, 2026).

[54] J. Cohen, "A coefficient of agreement for nominal scales," *Educ. Psychol. Meas.*, vol. 20, no. 1, pp. 37–46, 1960.

[55] C. D. Manning, P. Raghavan, and H. Schütze, *Introduction to Information Retrieval*. Cambridge, U.K.: Cambridge Univ. Press, 2008.

[56] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: Synthetic minority over-sampling technique," *J. Artif. Intell. Res.*, vol. 16, pp. 321–357, 2002.

[57] M. Grinberg, *Flask Web Development*, 2nd ed. Sebastopol, CA, USA: O'Reilly Media, 2018.

[58] *Systems and Software Engineering — Systems and Software Quality Requirements and Evaluation (SQuaRE) — System and Software Quality Models*, ISO/IEC 25010:2011, 2011.

[59] United Nations General Assembly, "Transforming our world: The 2030 Agenda for Sustainable Development," Resolution A/RES/70/1, Oct. 2015.

[60] J. Brooke, "SUS: A 'quick and dirty' usability scale," in *Usability Evaluation in Industry*, P. W. Jordan, B. Thomas, B. A. Weerdmeester, and I. L. McClelland, Eds. London, U.K.: Taylor & Francis, 1996, pp. 189–194.

[61] A. Bangor, P. T. Kortum, and J. T. Miller, "An empirical evaluation of the System Usability Scale," *Int. J. Hum.–Comput. Interact.*, vol. 24, no. 6, pp. 574–594, 2008.

GLOSSARY

Briefing. A consolidated chat message giving responders the incident type, urgency, statement, profile and map link.

Fail-safe delivery. The design rule that an alert is delivered to officers even if the AI components fail.

Haversine distance. The great-circle distance between two points on a sphere computed from latitude and longitude.

Hold-out test set. Data kept aside during training and used once to report final performance.

Incident type. The category of emergency described by a tourist's statement, for example theft, medical, harassment or fire.

Keyword fallback. A rule-based classifier (TYPE_KEYWORDS) used if the model bundle is unavailable and as a baseline.

Logistic Regression. A linear probabilistic classifier; the strongest incident-type candidate on the current corpus.

Macro-averaged F1. The mean of per-class F1-scores, giving each class equal weight.

Offline queue. Local storage of SOS payloads raised without connectivity, replayed when the browser is online again.

Random Forest. An ensemble of decision trees voting on the predicted class.

Responder route. The police, medical or fire channel selected by the responder Random Forest.

Safe Journey. An optional route-logging feature that attaches recent GPS points to an SOS briefing.

TF-IDF. A numeric weighting of words in a statement used as machine-learning input.

Tourist Police. The unit of the Sri Lanka Police that assists tourists.

Urgency. A priority level derived from the incident type; priority-one events trigger immediate dispatch.

Word error rate. The proportion of words in a transcript that are substituted, deleted or inserted relative to a reference.

APPENDICES

Appendix A: Metrics File and Corpus Support

The following structure shows the hold-out results recorded by the training script. Values marked in the text must be copied from the final metrics.json.

```
{
  "corpus": {"statements": 264, "classes": 14, "test_size": 0.2,
"random_state": 42},
  "incident_type": {
    "dummy":               {"macro_f1": 0.012},
    "keyword":             {"accuracy": 0.547, "macro_f1": 0.496},
    "random_forest":       {"accuracy": <insert>, "macro_f1": 0.359},
    "logistic_regression": {"accuracy": <insert>, "macro_f1": 0.707}
  },
  "responder": {
    "dummy":         {"macro_f1": 0.265},
    "random_forest": {"accuracy": 0.811, "macro_f1": 0.684}
  }
}
```

Table A.1: Incident types and corpus support (to be completed from Incident.VALID_TYPES)

No.	Incident type	Responder	Statements
1	Theft	Police	[n]
2	Medical	Medical	[n]
3	Harassment	Police	[n]
4	Fire	Fire	[n]
5	Road accident	Medical / Police	[n]
6	Unsafe location / lost	Police	[n]
7–14	[insert the remaining eight types]	[insert]	[n]

Appendix B: Retraining Procedure

Table A.2: Retraining procedure

Step	Action	Check
1	Add or edit statements in sos_ml_corpus.py	Each label is in Incident.VALID_TYPES
2	Run RETRAIN_SOS_ML.bat (train_sos_ml.py)	New metrics.json and sos_models.joblib written
3	Compare metrics with the previous run	No unexpected drop in responder or type macro-F1
4	Restart the unified back end (port 5002)	Classification source reported as random_forest, not fallback
5	Run tests/test_sos_ml.py	All tests pass

Appendix C: Simplified Code Excerpts

The excerpts show the structure of the implementation in simplified form [check against the repository before submission].

```
# Training (train_sos_ml.py / sos_ml_service.py), simplified
texts, types, responders = load_corpus()              # sos_ml_corpus.py,
264 rows
X_tr, X_te, y_tr, y_te, r_tr, r_te = train_test_split(
    texts, types, responders, test_size=0.2, stratify=types,
random_state=42)
vec = TfidfVectorizer(ngram_range=(1, 2), max_features=4000,
stop_words="english")
Xtr, Xte = vec.fit_transform(X_tr), vec.transform(X_te)   # fit on train
only
rf = RandomForestClassifier(n_estimators=180, class_weight="balanced",
random_state=42).fit(Xtr, y_tr)
lr = LogisticRegression(max_iter=2000, class_weight="balanced").fit(Xtr,
y_tr)
dum = DummyClassifier(strategy="most_frequent").fit(Xtr, y_tr)
```

```
# Runtime classification with fallback (ai_service.py), simplified
def classify_statement(text):
    clean = clean_statement(text)
    if BUNDLE is not None:
        X = BUNDLE["vectorizer"].transform([clean])
        probs = BUNDLE["type_model"].predict_proba(X)[0]
        i = probs.argmax()
        return dict(type=BUNDLE["type_model"].classes_[i],
confidence=float(probs[i]),
                    responder=BUNDLE["responder_model"].predict(X)[0],
source="random_forest")
    itype = keyword_classify(clean)
    return dict(type=itype, confidence=None,
responder=DEFAULT_RESPONDER[itype], source="keyword_fallback")
```

```
# Nearest station (routing.py), simplified
def nearest_station(lat, lon, responder):
    candidates = Station.query.filter_by(type=responder).all()
    return min(candidates, key=lambda s: haversine_km(lat, lon, s.lat,
s.lon))
```

Appendix D: System Usability Scale Items (Planned Instrument)

Participants of the planned usability study would rate each statement from 1 (strongly disagree) to 5 (strongly agree) [60].

1. I think that I would like to use this system frequently.
2. I found the system unnecessarily complex.
3. I thought the system was easy to use.

4. I think that I would need the support of a technical person to use this system.
5. I found the various functions in this system were well integrated.
6. I thought there was too much inconsistency in this system.
7. I would imagine that most people would learn to use this system very quickly.
8. I found the system very cumbersome to use.
9. I felt very confident using the system.
10. I needed to learn a lot of things before I could get going with this system.

Appendix E: Institutional Documents

Copies or descriptions of the letters connected with the Police Headquarters and immigration authorities, and the research proposal submitted to the Sri Lanka Police, are to be attached here [TBD: attach with the permission of the issuers; hide personal data].

Appendix F: Academic Integrity and Use of Generative AI

Generative AI tools were used to assist with drafting and structuring parts of the thesis text and with producing diagrams from descriptions supplied by the author. The training corpus is synthetic or paraphrased, and no real victim statements were used. The author remains responsible for the accuracy, originality, citations, code, data, analyses and conclusions. [The author must edit this statement to say exactly which tools were used and for which tasks, in line with the SLIIT policy, and confirm that all public sources cited were opened and checked.]